RMG-1: Relational Medium Gravity

Gravity as the low-energy relational geometry of a stable base-medium

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Purpose

RMG formulates gravity as the low-energy relational geometry of a stable base-medium.

Core sentence:

$$\boxed{\text{gravity is the low-energy relational geometry of a stable base-medium.}}$$

The starting object is a base-medium $\mathcal{M}$ with a stable low-energy phase. The tested GR/ADM structure is required to emerge as the low-energy fixed point of $\mathcal{M}$.

The protected low-energy phase admits only operational relational observables, not a separate local medium-frame field.

A central core gate is that all physical low-energy principal operators share one kinetic metric kernel.

A known no-go result is addressed at the outset. The Weinberg–Witten theorem forbids a massless spin-2 particle from emerging as a composite in a relativistic quantum field theory carrying a Lorentz-covariant conserved stress-energy tensor on a fixed background. RMG is not such a theory: it has no fixed background spacetime and no Lorentz-covariant stress-energy tensor defined on one. Spacetime, rods, clocks and the operational metric are themselves emergent relational records of $\mathcal{M}$, in the sense of Operational Relational Covariance. The background-independence of RMG is therefore the intended route around the Weinberg–Witten obstruction. This escape is necessary but not sufficient: the positive construction of exactly two transverse-traceless tensor radiation modes, with no extra physical scalar or vector gravitational mode, remains an open requirement carried by the record-sector factorization and spin-2 emergence gates.

Status labels

Every statement in RMG-1 carries one of the following labels: *Primitive Postulate* (minimal starting property of the base-medium); *Primitive Protection Principle* (low-energy relational rule excluding operators that violate the protection requirements); *Working Assumption* (temporary bridge requiring deeper derivation); *Result* (statement derived from listed assumptions); *Low-Energy Target* (structure required by observation and adopted as the low-energy fixed point); *Open Theorem* (claim still to be proven from $\mathcal{M}$); *Failure Gate* (condition whose failure invalidates the local core).

1 Core Terms

RMG (Relational Medium Gravity) is a gravitational framework in which a single base-medium $\mathcal{M}$ gives rise to matter modes, rods, clocks, signals and operational geometry. The deeper task is to explain why the universal low-energy geometry is the ADM/GR fixed point.

$\mathcal{M}$ is the base-medium; it is not assumed to be an ordinary scalar field. $\mathcal{W}$ is the relational medium-history. X_F is a foliation or embedding description of the same history $\mathcal{W}$.

$s(x)$ is the local operational factor shared by clocks, rods and signals in the weak-sector bridge description. u is the compression/lapse potential, with $s = e^{-u}$; it enters the action as a constraint multiplier, not as an independent propagating field.

$g^{\rm eff}_{\mu\nu}$ is the effective metric seen by low-energy matter modes. N, N^i, h_{ij} are the ADM lapse, shift and spatial metric variables in the low-energy target, and h^{TT}_{ij} is the transverse-traceless tensor/shear radiation sector. $u^\mu_{\mathcal{M}}$ denotes a putative independent local medium-frame field; it must not appear as an independent physical field in the tested RMG core.

$Q^{\mu\nu}_{AB}$ is the highest-derivative low-energy kinetic response matrix for perturbations of matter and geometry modes. $G^{\mu\nu}$ is the common principal metric kernel, identified with $g^{\mu\nu}_{\rm eff}$ if the factorization gate succeeds. K_{AB} is an internal/species kinetic matrix: it carries allowed differences between particle families, tail shapes, polarizations, phases and internal geometries. $B^{\mu\nu}_{AB}$ is the non-factorizing failure matrix; it must vanish on physical low-energy principal modes, up to gauge, internal or suppressed correction terms.

$\Delta_{\mathcal{M}}$ is the residual medium contribution to the low-energy Hamiltonian/constraint algebra after the ADM/EH target terms are separated; the tested core requires $\Delta_{\mathcal{M}} = 0$, or suppression outside all tested regimes.

$\mathcal{R}$ is the record sector: low-energy modes that can participate in operational measurement as rods, clocks, detector components, memory or record components, exchanged signals or stable external matter excitations. $\mathcal{I}$ is the internal sector: confined internal modes, gauge directions, constraint variables, off-shell auxiliary modes, short-range internal responses and high-energy modes outside the tested local regime. Internal modes may contribute to mass, binding energy, charge, spin, colour, tail shape, effective couplings and total energy, but must not define a separate external spacetime cone in the tested local core.

2 Primitive Postulates

Primitive Postulate 1 (One base-medium)**.** *All low-energy matter modes, rods, clocks, signals and geometric excitations are collective modes of one base-medium $\mathcal{M}$.*

Remark (mathematical status of $\mathcal{M}$). This postulate fixes only the ontological commitment of RMG-1: one substrate whose low-energy observable structures are collective or coarse-grained modes. It does not assume that $\mathcal{M}$ is an ordinary scalar field, nor does it by itself specify a state space, action, symplectic structure, internal redundancy group or record observables. Those data belong to the later base-medium input package and candidate order-parameter construction. The postulate therefore asserts existence and exhaustiveness; the microscopic content remains an open construction target.

Primitive Postulate 2 (Stable low-energy phase)**.** *The base-medium has a stable low-energy phase in which matter modes, rods, clocks, signals and operational geometry are well-defined.*

Primitive Postulate 3 (Minimal operational node scale)**.** *The base-medium as one ontological object has no ordinary microscopic radius. A length becomes physical only when the base-medium is distinguishable as recordable nodes. The first distinguishable node/record scale is denoted*

$$\ell_*.$$

Below ℓ_*, two nodes are not yet separate operational objects: there is no independent distance, delay, clock comparison or mass comparison. Above ℓ_*, medium dilution also rescales rods, clocks *and the visible node scale. Thus ℓ_* is an operational closure scale, not the radius of a mechanical particle.*

In the present RMG-1 core, ℓ_* is identified with the observed Planck-order minimal record scale. This is not a second medium and not a species-dependent parameter; it is the universal first readable separation of the one base-medium.

Primitive Postulate 4 (Operational Relational Covariance)**.** *Local low-energy observables are invariant under re-embedding of the same relational medium-history:*

$$\mathcal{W} \sim X_F(\mathbb{R} \times \Sigma) \sim X_{F'}(\mathbb{R} \times \Sigma).$$

Thus coordinates, slicings and internal clock choices are descriptions, not extra physical structures. In particular, no independent local preferred-frame field $u^\mu_{\mathcal{M}}$ appears in the tested core.

Remark (operational content of ORC). The re-embedding statement above has three explicit operational consequences that load the closure of the gravitational core:

1. Substrate-internal variables that do not enter operational record comparisons are gauge / bookkeeping, not physical observables. In particular, no hidden substrate clock, no preferred rest frame, and no additional substrate-internal label distinguish records.

2. All stable record-forming matter reads the same local operational geometry. No substrate-internal mode produces a distinguishable second operational time, second operational length, or second principal cone.

3. Background uniform substrate states (e.g. the uniform phase advance $\langle\theta\rangle = \nu_0 t$ on the stable phase $\mathcal{M}_0$) are operationally hidden; only the fluctuations on top of such backgrounds are physical. Only fluctuations relative to the background carry operational content; this is the relational form of Galilean and Lorentz invariance.

ORC is the principal axiom that closes the three local-gravity roots (Root A: one carrier; Root B: ADM/EH closure; Root C: total-energy source) into derived results. ORC itself is taken here as a primitive postulate; its derivation from a deeper micro principle remains a long-term research target.

Primitive Postulate 5 (Positive physical energy)**.** *Physical excitations of $\mathcal{M}$ have Hamiltonian bounded below:*

$$H - H_{\text{vac}} \geq 0.$$

All physical degrees of freedom carry positive kinetic energy. The trace/volume direction of the DeWitt supermetric belongs to the constrained sector and does not propagate as an independent mode.

Primitive Postulate 6 (Local effective hierarchy)**.** *The low-energy theory is local and organized by a derivative or scale expansion:*

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{core}} + \ell^2 \mathcal{L}_4 + \ell^4 \mathcal{L}_6 + \cdots .$$

Corrections may exist, but they must be suppressed or forbidden in tested local regimes.

3 Operational Bridge

This section gives the minimal bridge needed to recover the weak gravitational sector; the microscopic derivation is treated separately.

The universal factor $s(x)$ is a bridge variable, not a primitive object. Its deeper origin is the universal principal symbol of all low-energy matter modes.

Working Assumption 1 (Universal operational factor)**.** *In the slowly varying low-energy regime, all stable matter modes share one local operational factor:*

$$s = s(x).$$

Equivalently, all matter modes see one effective metric $g_{\mu\nu}^{\text{eff}}$.

Working Assumption 2 (Reciprocal clock/rod response)**.** *Local clocks and rods respond as:*

$$d\tau = sdt, \qquad d\ell = s^{-1}dx.$$

Result 1 (Square signal law)**.** *If the local physical signal speed is $c = d\ell/d\tau$, then:*

$$c = \frac{s^{-1}dx}{sdt} = s^{-2}\frac{dx}{dt}.$$

Therefore:

$$\boxed{\frac{c_{\text{coord}}}{c} = s^2.}$$

Result 2 (Coherent endpoint-counting route to the square law)**.** *The reciprocal clock/rod bridge is not produced by a longer courier path. A longer path at fixed carrier speed gives only a linear delay. A square response is obtained when the stable record sector is controlled by one coherent base-medium amplitude*

$$\boxed{\rho := |\Phi_{\mathcal{M}}| = \sqrt{n}}$$

and the local record coefficients scale as

$$\boxed{C_t : C_0 : C_x = 1 : \rho : \rho^2.}$$

Here C_t is the common-rhythm time coefficient, C_0 is the local stable gap coefficient and C_x is the spatial link-stiffness coefficient. Define

$$\boxed{s := \rho^{1/2} = n^{1/4}.}$$

Then

$$C_t : C_0 : C_x = 1 : s^2 : s^4.$$

For a stable quadratic record mode,

$$\omega_0 \propto \sqrt{\frac{C_0}{C_t}} = s, \qquad \ell_{\text{coord}} \propto \sqrt{\frac{C_x}{C_0}} = s, \qquad c_{\text{coord}} \propto \sqrt{\frac{C_x}{C_t}} = s^2 c.$$

Thus the coherent endpoint-counting completion derives

$$\boxed{d\tau = sdt, \qquad d\ell = s^{-1}dx, \qquad \frac{c_{\text{coord}}}{c} = s^2.}$$

The exponent origin is:

$$\boxed{\textit{site/local gap} \sim \rho, \qquad \textit{link/spatial stiffness} \sim \rho_a\rho_b \to \rho^2.}$$

Working Assumption 3 (Additive compression and multiplicative response)**.** *Weak compression potentials add:*

$$u_{12} = u_1 + u_2.$$

Process factors multiply:

$$s_{12} = s_1 s_2.$$

Result 3 (Exponential layering)**.** *The continuous solution with weak-field normalization is:*

$$\boxed{s = e^{-u}.}$$

Result 4 (ADM lapse identification)**.** *For a normal ADM observer with zero shift displacement,*

$$d\tau = N dt.$$

The RMG bridge clock response gives:

$$d\tau = s dt.$$

Therefore, inside the operational bridge,

$$\boxed{N = s = e^{-u}.}$$

The remaining task is to derive the universal factor s and the exponential layering from $\mathcal{M}$.

Result 5 (Weak operational metric)**.** *The bridge metric is:*

$$\boxed{ds^2 = -e^{-2u} c^2 dt^2 + e^{2u} \delta_{ij} dx^i dx^j.}$$

4 One-Kernel Factorization Gate

The universal matter metric must derive from a common highest-derivative kinetic kernel. For low-energy perturbations $\delta\Phi^A$, denote by $Q^{\mu\nu}_{AB}$ the principal kinetic response. For a covector k_μ, define the principal symbol:

$$\boxed{P_{AB}(k) = Q^{\mu\nu}_{AB} k_\mu k_\nu.}$$

Let $\Pi_{\mathcal{R}}$ project onto the physical record subspace after gauge and constraint directions are removed.

The labels A, B range over every stable record-forming excitation relevant to the local core: elementary medium excitations, effective matter modes, photons/signals, composite records such as atoms, rods and clocks, and detector components. Modes outside this class may be internal or confined, but they must not act as external spacetime carriers.

The principal symbol is extracted in the relevant low-energy effective regime, on the physical on-shell branch after gauge and constrained directions are removed. It is not a blind UV limit outside the record theory:

$$\boxed{k \ll \Lambda_{\mathrm{UV}}, \qquad \delta\Phi \in \Pi_{\mathcal{R}}, \qquad \text{dispersion evaluated on a physical record branch.}}$$

Open Theorem 1 (Factorization gate)**.** *Derive from the protected base-medium phase:*

$$\boxed{Q^{\mu\nu}_{AB} = G^{\mu\nu} K_{AB}.}$$

Then all physical low-energy modes share one principal metric:

$$\boxed{G^{\mu\nu} = g^{\mu\nu}_{\mathrm{eff}}.}$$

Common characteristic cones do not suffice: the physical principal symbol must also be free of unsuppressed species or polarization birefringence, and a single base medium does not by itself prevent internal or confined modes from defining separate cones. The RMG statement is therefore restricted:

$$\boxed{\text{one stable record-forming low-energy phase} \Rightarrow \text{one operational spacetime cone for records.}}$$

Modes outside the record sector may have different internal dispersions, but they must not define separate external rods, clocks, signals or spacetime metrics. The pass condition reads:

$$\boxed{\Pi_{\mathcal{R}} P(k) \Pi_{\mathcal{R}} = (G^{\mu\nu} k_\mu k_\nu)\, K_{\mathcal{R}}.}$$

Allowed remainders are gauge directions, constraint variables, internal algebra, lower-derivative terms or suppressed corrections. The local core fails if the physical principal symbol contains:

$$\boxed{P^{\text{phys}}_{AB}(k) = (G^{\mu\nu} k_\mu k_\nu)\, K_{AB} + B_{AB}(k), \qquad B_{AB}(k) \neq 0,}$$

where B_{AB} produces an observable species-private cone, polarization splitting, preferred-frame tensor or clock-dependent redshift.

Why this is non-trivial in RMG. In GR, one metric for all matter is usually imposed by minimal coupling. Root A is not allowed to assume that result. It must derive why all stable record-forming modes read one external cone from the base-medium spectrum and record projection. If this fails, later ADM closure or source universality cannot repair the local core.

Open Theorem 2 (Record-sector no-birefringence principle)**.** *All stable modes that can form local records, rods, clocks and exchanged signals must belong to one operational record sector. In that sector:*

$$\boxed{B^{\text{phys}}_{AB}(k) = 0}$$

for unsuppressed principal spacetime response. Internal differences may remain in K_{AB}, masses, charges, spin, color, topology, phase and tail geometry; they must not appear as separate external spacetime cones. This is a protection principle in RMG-1 and still requires derivation from $\mathcal{M}$.

Open Theorem 3 (Record-phase route to factorization)**.** *Derive the chain:*

$$\boxed{\textit{stable record phase} \Rightarrow \textit{common operational records} \Rightarrow \textit{no observable species-private spacetime cone} \Rightarrow \textit{no physical}}$$

This chain remains to be derived from $\mathcal{M}$.

Open Theorem 4 (One-tensor factorization lemma)**.** *For any candidate common record tensor $G^{\mu\nu}$, decompose the principal kinetic matrix as:*

$$\boxed{Q^{\mu\nu}_{AB} = G^{\mu\nu} K_{AB} + B^{\mu\nu}_{AB}.}$$

If the stable record phase contains only one observable rank-two spacetime principal tensor, then the non-factorizing part $B^{\mu\nu}_{AB}$ has no unsuppressed physical projection on external record-forming modes:

$$\boxed{\Pi_{\mathcal{R}} B^{\mu\nu}_{AB} \Pi_{\mathcal{R}} = 0}$$

up to gauge, confined, gapped, damped, unstable or high-energy suppressed pieces. Otherwise $B^{\mu\nu}_{AB}$ would define a second observable cone, anisotropy, preferred-frame tensor or species-private clock response. This is an open lemma: RMG must still derive the one-observable-tensor premise from the base medium.

Gate 1 (Observable tensor basis gate). *In the tested local record core, the physical principal symbol must contain no unsuppressed observable rank-two tensor other than the common record tensor* $G^{\mu\nu}$. *Equivalently, any decomposition*

$$\boxed{Q_{AB}^{\mu\nu} = G^{\mu\nu} K_{AB} + B_{AB}^{\mu\nu}}$$

must satisfy

$$\boxed{\Pi_{\mathcal{R}} B_{AB}^{\mu\nu} \Pi_{\mathcal{R}} = 0}$$

at leading principal order, except for gauge, constraint, internal/confined, gapped, damped, unstable, high-energy suppressed or explicitly large-scale correction sectors. Forbidden local-core pieces include unsuppressed

$$\boxed{U^\mu U^\nu, \quad v^\mu v^\nu, \quad \sigma^{\mu\nu}, \quad \omega^\mu{}_\alpha \omega^{\alpha\nu}, \quad \nabla^\mu u \nabla^\nu u, \quad S_A^{\mu\nu}.}$$

If any such tensor survives in physical record propagation, RMG predicts a preferred frame, anisotropic cone, species-private metric or birefringence. This gate is open from $\mathcal{M}$.

Open Theorem 5 (Master gate: no-birefringent record carrier). *The master local-core gate is:*

$$\boxed{\textit{all low-energy record-forming tails project onto one common carrier}}$$

so that the physical record-sector principal symbol has no unsuppressed species or polarization birefringence:

$$\boxed{P_{AB}^{\text{phys}}(k) = (G^{\mu\nu} k_\mu k_\nu)\, K_{AB}.}$$

If this gate closes, then one operational cone, one universal clock factor $s(x)$, *the ADM lapse bridge* $N = s = e^{-u}$, *the total-energy source gate and the compact internal-energy/no-external-cone split become linked consequences of the same record-sector protection.*

Here "one common carrier" means one shared base-medium vibration channel, not one elementary scalar field and not one tail substance per particle. Different sources may imprint different local patterns on the same medium; the tested local core allows only one external record carrier/cone.

Open Theorem 6 (Stable-tail common-carrier route). *For every stable record-forming oscillon or matter mode* A, *decompose its long tail as:*

$$\boxed{T_A = \Pi_{\text{common}} T_A + \Pi_{\text{internal}} T_A.}$$

The theorem target is:

$$\boxed{\text{principal}(\Pi_{\text{common}} T_A) = G^{\mu\nu} k_\mu k_\nu \quad \textit{for all record-forming } A,}$$

while $\Pi_{\text{internal}} T_A$ *may carry source-specific frequency, phase, spin, charge, topology, multipoles and short-range structure, but must not define a separate external spacetime cone.* T_A *is a source imprint in the same base-medium vibration, not a separate long-range field assigned to species* A. *Modes whose tails cannot project onto the common carrier are internal, confined, gapped, constrained or non-record-forming.*

Open Theorem 7 (Universal operational factor from one cone). *For a clock family* A, *write:*

$$d\tau_A = s_A(x) dt.$$

If the one-kernel factorization gate holds, then all record-forming clock families read the same spacetime kernel $G^{\mu\nu}$. Hence:

$$\boxed{s_A(x) = C_A\, s(x)\,[1 + \epsilon_A(x)].}$$

Here C_A is a constant internal clock calibration, while $\epsilon_A(x)$ is the species-dependent environmental response that would violate the universality requirement. The tested core requires:

$$\boxed{\epsilon_A(x) = 0}$$

or suppression outside tested regimes. After constant calibration,

$$\boxed{s_A(x) = s_B(x) = s(x).}$$

Equivalently,

$$\boxed{\nabla_\mu \ln \frac{s_A}{s_B} = 0}$$

up to suppressed corrections. If this fails, the failure is the clock-level version of a nonzero $B^{\mu\nu}_{AB}$.

The exact target is $\epsilon_A = 0$. If a candidate leaves a residual ϵ_A, it must be bounded against clock-comparison, redshift, composition-dependence and species-dependent signal-cone tests; it cannot remain an unconstrained symbolic remainder.

Different oscillons may excite the same base-medium with different local frequencies, phases, polarizations and internal geometries. The factorization requirement constrains only the external record carrier of their long-range tails, which must be the same base-medium vibration channel:

$$\boxed{\text{diverse local imprints, one base-medium, one operational principal cone.}}$$

The possible failure is:

$$\boxed{Q^{\mu\nu}_{AB} = G^{\mu\nu} K_{AB} + B^{\mu\nu}_{AB}, \qquad B^{\mu\nu}_{AB} \neq 0.}$$

In the tested core, $B^{\mu\nu}_{AB}$ must not contain an independent preferred-frame tensor, species-private metric, or geometry-sector cone different from the matter cone.

5 Tail Superposition And Projection Gate

Different oscillons may produce different local tail imprints in the same base-medium. Gravity does not read the full microscopic imprint. It reads the common pressure/refraction projection:

$$\boxed{\Pi_{\text{grav}}(T_A) = q_A \mathcal{G}.}$$

Here T_A is the source imprint associated with particle family A, $\mathcal{G}$ is the common base-medium pressure carrier, and q_A is the pressure-deficit strength. T_A must not be read as a separate long-range field or separate spacetime channel.

Open Theorem 8 (Universal gravitational projection)**.** *For all stable low-energy record-forming matter modes, derive:*

$$\boxed{q_A \propto E_A/c^2.}$$

Here E_A means total physical energy of the source mode or composite system:

$$\boxed{E_A = E_{\text{rest}} + E_{\text{kinetic}} + E_{\text{binding}} + E_{\text{internal}}.}$$

Confined/internal modes do not need their own free long gravitational tails; they contribute through the total energy of the composite external tail.

Then many different tails source one pressure profile:

$$\boxed{u(\mathbf{x}) = \sum_i u_i(\mathbf{x}), \qquad \nabla^2 u \sim -\rho_E}$$

in the weak-field regime, and the same u determines the effective geometry.

Equivalently, the far long-range pressure tail must have an energy monopole:

$$\boxed{u_A(r,\Omega) = \frac{Q_A^{(0)}}{r} + O(r^{-2}), \qquad Q_A^{(0)} = \eta\, E_A/c^2.}$$

The theorem target is that the common pressure source density equals total physical energy density in mass units:

$$\boxed{\sigma_{\text{grav}} = \alpha_E\, \rho_E/c^2.}$$

This is expected because the long common pressure channel should couple to the conserved scalar generator of stable medium evolution, namely total energy.

A more specific source theorem target is:

$$\boxed{S_u(x) := \frac{1}{N\sqrt{h}}\frac{\delta S_{\text{m}}}{\delta u(x)} = \rho_E(x) + \text{suppressed corrections.}}$$

Here S_u is the source seen by the compression/lapse variable u. The reason is that u changes the local operational process rate, and the generator of local process-rate change is the Hamiltonian density. Thus the common pressure channel should read $\rho_E = \mathcal{H}_{\text{phys}}$, not a species-private tail detail.

Species-specific tail structure may remain in K_{AB} or in internal sectors (spin, charge, colour, phase, polarization, topology), but must not create a second long-range spacetime carrier. If an unsuppressed species-private source term remains in S_u, the equivalence-principle source gate fails.

Open Theorem 9 (Matter-coupled closure and source universality)**.** *Beyond vacuum ADM closure, the matter-coupled constraints must close as well:*

$$\boxed{\mathcal{H}_{\text{tot}} = \mathcal{H}_{\text{grav}} + \mathcal{H}_{\text{matter}} \approx 0, \qquad \mathcal{H}_{i,\text{tot}} = \mathcal{H}_{i,\text{grav}} + \mathcal{H}_{i,\text{matter}} \approx 0.}$$

The closure target is:

$$\boxed{\{H_{\text{tot}}[N], H_{\text{tot}}[M]\} = D_{\text{tot}}\left[h^{ij}(N\partial_j M - M\partial_j N)\right].}$$

This requires stable matter families to use the same record metric and the same total-energy source:

$$\boxed{S_{\text{matter}} = S_{\text{matter}}[g,\Psi], \qquad S_u = \rho_E.}$$

No unsuppressed species-private gravitational source may remain:

$$\boxed{\rho_{\text{grav}} \neq \rho_E + \sigma_{\text{species}} \quad \textit{with } \sigma_{\text{species}} \textit{ physical and unsuppressed.}}$$

Internal/confined modes, including binding and gluon-like internal energy, must contribute to E_{total} while not defining an independent external spacetime carrier. This gate is open from $\mathcal{M}$.

For the ADM matter action

$$\boxed{S_{\mathrm{m}} = \int dt\, d^3x \left(\pi_A \dot{\psi}_A - N\mathcal{H}_{\mathrm{m}} - N^i \mathcal{H}_i\right), \qquad N = e^{-u},}$$

variation at fixed h_{ij} gives

$$\boxed{\delta_u S_{\mathrm{m}} = \int dt\, d^3x\, N\mathcal{H}_{\mathrm{m}}\, \delta u, \qquad \frac{1}{N\sqrt{h}} \frac{\delta S_{\mathrm{m}}}{\delta u} = \rho_E.}$$

Thus the low-energy ADM bridge already identifies the u-source with the Hamiltonian density. It remains to derive $N = e^{-u}$ and the universal ADM bridge from the base-medium.

6 Delta-M Candidate Theorem

Open Theorem 10 (Conditional Delta-M theorem)**.** *If the protected low-energy phase has:*

$$\boxed{\textit{one stable operational record phase with a common carrier/principal cone,}}$$

$$\boxed{\textit{exact Operational Relational Covariance,}}$$

$$\boxed{Q^{\mu\nu}_{AB} = G^{\mu\nu} K_{AB},}$$

and a local two-derivative low-energy hierarchy, then the tested core permits no unsuppressed independent medium residue:

$$\boxed{\Delta_{\mathcal{M}} = 0}$$

up to suppressed high-energy or large-scale correction windows.

7 Static Weak-Field Sector

Working Assumption 4 (Hamiltonian source)**.** *In the weak static regime, the compression/lapse constraint is sourced by total physical energy density ρ_E:*

$$\rho_E(\mathbf{x}) \geq 0.$$

The inertial mass of a localized source is:

$$M = \frac{1}{c^2} \int d^3x\, \rho_E(\mathbf{x}).$$

Working Assumption 5 (Base-node calibration of G)**.** *The source coefficient in the weak static sector is fixed by the minimal operational node scale:*

$$\boxed{G_{\mathrm{RMG}} = \frac{c^3 \ell_*^2}{\hbar}.}$$

Equivalently, the gravitational stiffness is

$$\boxed{\frac{c^4}{4\pi G_{\mathrm{RMG}}} = \frac{\hbar c}{4\pi \ell_*^2}.}$$

When the universal operational node scale is identified with the observed Planck-order length, G_{RMG} equals the measured Newton constant. The remaining deeper task is not to choose a smaller length, but to derive ℓ_* itself from the stable base-medium record dynamics.

Below, G denotes this calibrated $G_{\rm RMG}$ unless explicitly stated.

Use the static energy functional:

$$\mathcal{E}[u] = \frac{c^4}{8\pi G_{\rm RMG}} \int d^3x\, |\nabla u|^2 - \int d^3x\, \rho_E u.$$

Result 6 (Static source equation)**.** *Varying u gives:*

$$\boxed{\nabla^2 u = -\frac{4\pi G_{\rm RMG}}{c^4}\rho_E.}$$

With $u(\infty) = 0$:

$$u(\mathbf{x}) = \frac{G_{\rm RMG}}{c^4} \int d^3x' \, \frac{\rho_E(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|}.$$

Far from the source:

$$\boxed{u(r) = \frac{G_{\rm RMG} M}{rc^2} + O(r^{-2}).}$$

Thus the gravitational charge is inertial mass because the source is total energy.

8 PPN Weak-Field Check

Using:

$$U = c^2 u, \qquad U(r) = \frac{G_{\rm RMG} M}{r},$$

the metric gives:

$$g_{00} = -e^{-2u} = -1 + 2u - 2u^2 + O(u^3),$$
$$g_{ij} = e^{2u}\delta_{ij} = (1 + 2u + O(u^2))\delta_{ij}.$$

Therefore:

$$\boxed{\gamma = 1, \qquad \beta = 1.}$$

9 Constraint And Radiation Targets

The low-energy dynamic description must be expressible in ADM variables:

$$ds^2 = -N^2 c^2 dt^2 + h_{ij}(dx^i + N^i dt)(dx^j + N^j dt).$$

Weak split:

$$N = 1 - u + O(2),$$
$$h_{ij} = (1 + 2u)\delta_{ij} + h_{ij}^{TT} + O(2),$$
$$N_i = \mathcal{A}_i + O(2).$$

Low-Energy Target 1 (Scalar and vector sectors)**.** *The scalar/compression variable and the vector/flow variable must be constraint or near-zone variables at low energy, not independent vacuum radiation modes.*

The absence of scalar/vector radiation is protected only if lapse and shift remain non-propagating constraint variables.

Target form:

$$\nabla^2 u = -4\pi G\rho/c^2,$$
$$\nabla^2 \mathcal{A}_i = -\kappa_V G\mathcal{P}_i^T/c^3.$$

Low-Energy Target 2 (Tensor/shear radiation)**.** *The only propagating vacuum gravitational modes at low energy must be:*

$$h_+, \qquad h_\times.$$

They must have positive energy and propagate on the single record cone:

$$c_{\mathrm{GW}} = c.$$

Here c is not one of several speeds tuned to agree; it is the unique local signal speed defined by the common record cone after Root A closes.

Target wave equation:

$$\left(\frac{1}{c^2}\partial_t^2 - \nabla^2\right) h_{ij}^{TT} = \kappa_T G T_{ij}^{TT}/c^4.$$

Vacuum:

$$\left(\frac{1}{c^2}\partial_t^2 - \nabla^2\right) h_{ij}^{TT} = 0.$$

The propagation target also excludes massive vector modes, massive spin-2 leakage, a second metric tensor mode, and any extra shear channel in the tested local sector. Propagation alone is not sufficient: compact-binary emission must use the same protected tensor channel, with no leading scalar-dipole or scalar-monopole excess in binary-pulsar or waveform tests.

Gate 2 (GW detection: common-mode cancellation versus shear signal)**.** *A uniform or adiabatic compression of the local record system rescales the apparatus, clocks, rulers and light standards together; the scalar/compression channel u therefore couples to an interferometer signal only through differential terms. The local Michelson-like difference channel projects out the common-mode part:*

$$\boxed{\Pi_{\mathrm{det}}\,\delta g_{\mathrm{common}} = 0.}$$

The detector-visible gravitational-wave channel must instead be a differential record deformation:

$$\boxed{\Pi_{\mathrm{det}}\,\delta g_{\mathrm{rec}} = \Pi_{TT}\,\delta g_{\mathrm{rec}} + \Pi_{\mathrm{shear/defect}}\,\delta u_{\mathrm{noncommon}}.}$$

The first term is the protected tensor/shear sector h_{ij}^{TT}. The second term is allowed only as a controlled conversion of breathing/compression into detector-visible shear, tidal or symmetry-defect response. It must not survive as an unsuppressed local scalar radiation mode.

Here "defect projection" means a detector-visible differential record deformation which has a TT/shear component after projection onto the detector response. It is a placeholder for a later detector-response calculation, not a license to re-label a pure scalar breathing mode as a tensor wave.

A local apparatus cannot register a uniform rescaling of the medium from which its rods, clocks and light standards are built; it can register only anisotropic or differential record deformation. Observed gravitational-wave signals must therefore be carried by the protected TT/shear response or by a suppressed noncommon conversion channel, not by a pure common-mode scalar breathing.

Open Theorem 11 (Constraint protection route)**.** *Derive from the base-medium that the local low-energy variables enter in ADM-like form:*

$$\boxed{S_{\text{local}} = \int dt\, d^3x \left(\pi^{ij}\dot{h}_{ij} - N\mathcal{H} - N^i\mathcal{H}_i\right)}$$

with no independent low-energy kinetic terms

$$\boxed{\dot{N}^2, \qquad \dot{N}_i\dot{N}^i, \qquad \dot{u}^2, \qquad \dot{\mathcal{A}}_i\dot{\mathcal{A}}^i}$$

for physical scalar/vector gravitational variables. If N and N^i impose first-class Hamiltonian and momentum constraints which close under time evolution, then u and local flow/shift variables are constraint/gauge or near-zone variables, not independent radiation. The vacuum gravitational record sector then has only

$$\boxed{h_+, \qquad h_\times}$$

as physical propagating modes. This is the local mechanism behind flow decoupling and the preferred-frame null target.

10 ADM/EH Fixed Point

Low-Energy Target 3 (Nonlinear low-energy completion)**.** *The nonlinear low-energy fixed point is:*

$$\boxed{S_{\text{low}} = \frac{c^4}{16\pi G}\int d^4x\sqrt{-g_{\text{rec}}}\,(R[g_{\text{rec}}] - 2\Lambda) + S_{\text{matter}}[g_{\text{rec}}, \Psi] + S_{\text{suppressed}}.}$$

Coefficient status. In the ADM filter below, α is a temporary curvature coefficient. The Einstein–Hilbert target fixes $\alpha = 1$ after the normalization of the measured Newton constant G is chosen. The cosmological term Λ is kept as a local-core background parameter in the MVP, but in the full program it must be derived, regulated, or absorbed into the cosmological-background closure. It may not be used to repair a local ADM/EH or source-universality failure.

ADM form:

$$S_{\text{ADM}} = \int dt\, d^3x \left(\pi^{ij}\dot{h}_{ij} - N\mathcal{H} - N^i\mathcal{H}_i\right).$$

Hamiltonian and momentum constraints:

$$\mathcal{H} \approx 0, \qquad \mathcal{H}_i \approx 0.$$

The target constraint algebra includes:

$$\{H[N], H[M]\} = D\left[h^{ij}(N\partial_j M - M\partial_j N)\right].$$

Operational Relational Covariance makes ADM closure plausible, but it does not replace the proof.

Open Theorem 12 (Main closure theorem)**.** *Derive from the base-medium:*

$$\boxed{\mathcal{M} \Rightarrow \textit{universal principal symbol} \Rightarrow \textit{ADM constraint closure} \Rightarrow S_{\text{EH}} + S_{\text{matter}}.}$$

Open Theorem 13 (Hypersurface-deformation closure filter). *The local Hamiltonian constraint must pass the ADM deformation-algebra filter. For the two-derivative ansatz*

$$\boxed{\mathcal{H} = \frac{1}{\sqrt{h}}\left(\pi^{ij}\pi_{ij} - \lambda\pi^2\right) - \sqrt{h}(\alpha R - 2\Lambda) + \Delta_{\mathcal{M}},}$$

the target bracket is:

$$\boxed{\{H[N], H[M]\} = D\big[h^{ij}(N\partial_j M - M\partial_j N)\big]\,.}$$

This filter requires, in the protected local core:

$$\boxed{\lambda = \frac{1}{2}, \qquad \Delta_{\mathcal{M}} = 0 \quad \textit{or constrained/gauge/suppressed},}$$

and no unsuppressed higher-derivative curvature terms at the tested two-derivative fixed point. If this closure filter is derived from $\mathcal{M}$, the trace/volume direction of the DeWitt supermetric belongs to the constrained sector and does not propagate as an independent scalar mode, and the local two-derivative fixed point is $S_{\text{EH}} + S_{\text{matter}}$ up to suppressed corrections.

Open Theorem 14 (Constraint preservation and matter-coupled Dirac closure). *ADM form is not enough. The Hamiltonian and momentum constraints must be preserved by the evolution they generate:*

$$\boxed{\dot{\mathcal{H}} \approx 0, \qquad \dot{\mathcal{H}}_i \approx 0.}$$

The matter-coupled Dirac analysis must show that primary constraints lead to the Hamiltonian and momentum constraints, that secondary constraints close without making lapse or shift physical, and that any second-class remnants are solved or remove only nonphysical variables. If matter coupling turns the first-class ADM system into a non-closing algebra, the local target action fails even if the vacuum bracket has the desired form.

11 Local Gravity Closure Package

The many open local-core gates reduce to three root theorems.

Open Theorem 15 (Root A: one external record carrier). *Derive from the base-medium stable record phase:*

$$\boxed{\textit{stable external record-forming mode} \Rightarrow \textit{one common external long-tail carrier channel.}}$$

All non-common branches must be internal, confined, gapped, damped, constrained, gauge or unstable. If Root A closes, it gives one operational principal cone, no low-energy birefringence and the factorization

$$\boxed{Q^{\mu\nu}_{AB} = G^{\mu\nu}K_{AB}}$$

on physical record modes.

Open Theorem 16 (Root A spectral/resonance closure). *Root A must be proven by a spectral statement around the stable record phase $\mathcal{M}_0$. Linearize the base-medium dynamics:*

$$\boxed{\delta\mathcal{E}[\mathcal{M}_0] = \mathcal{L}_{\mathcal{M}_0}\delta\Phi + \cdots\,.}$$

The required split is:

$$\boxed{\mathrm{Spec}(\mathcal{L}_{\mathcal{M}_0}) = \mathcal{B}_{\mathrm{rec}} \cup \mathcal{B}_{\mathrm{int}}, \qquad N_{\mathrm{cone}}(\mathcal{B}_{\mathrm{rec}}^{\mathrm{gapless}}) = 1}$$

as one external record carrier channel. This is a one-common-carrier condition, not a one-scalar-mode condition. The common channel may contain shared polarization, shear or tensor patterns, but it must not split into species-private external carriers. Stable record-forming tail imprints must phase-lock to the base-medium rhythm ν_0 or its allowed harmonic channel:

$$\boxed{\Pi_{\mathrm{rec}} T_A^{\mathrm{long}} = Z_A \mathcal{C} + \textit{internal/lower-order details}, \qquad \mathrm{principal}(\mathcal{C}) = G^{\mu\nu} k_\mu k_\nu.}$$

All other branches must be internal, confined, gapped, damped, constrained, gauge, unstable, high-energy suppressed or large-scale-only. This is the microscopic calculation still needed for Root A.

Suppression catalogue. *These escape categories are not free passes. Each requires a structural or quantitative reason why it is absent from local record observables:*

- *high-energy suppressed: corrections scale with E/Λ_{UV} or k/Λ_{UV} and are below laboratory, Solar-System and GW bounds;*
- *large-scale-only: the term activates only outside the local tested regime and does not affect Solar-System, binary-pulsar or local GW tests;*
- *confined or internal: the mode cannot propagate as an external rod, clock, signal or detector record;*
- *gapped or damped: the mode has no coherent long-range low-energy propagation in the tested regime;*
- *gauge or constrained: the mode is removed by $\Omega_{\mathcal{M}}$, a first-class generator or an elliptic/near-zone constraint;*
- *unstable: the branch is excluded from the stable phase or decays before record formation.*

If no such suppression or removal test is available, the branch remains a Root A fail-risk rather than an allowed remainder.

Root A–Root B compatibility. Root A must not assume ADM closure, but its record/internal projectors and the gauge/constraint reduction induced by $\Omega_{\mathcal{M}}$ must remain compatible with the canonical analysis needed in Root B. A one-cone spectrum obtained by a projection that destroys the later ADM phase space is not a valid Root A pass.

Remark (Root A closure status under ORC). Conditional on the Operational Relational Covariance postulate of Section "Primitive Postulates", the Root A open theorem closes as a derived Result. The argument proceeds in three steps: (i) Goldstone–Bogoliubov spectral counting on the stable phase $\mathcal{M}_0$ with broken $U(1)_\nu$ rhythm yields one type-I scalar Goldstone (the phonon) plus two transverse-traceless tensor modes; (ii) the "one operational geometry" clause of ORC forces $c_{\mathrm{phase}} = c_{\mathrm{graviton}}$, so all gapless record modes share the single principal symbol $G^{\mu\nu} k_\mu k_\nu$; (iii) stable record-forming oscillons project on this single channel with species-dependent amplitudes Z_A but a common cone. Without ORC the closure remains open; the microscopic derivation of ORC itself is left as the deeper open program.

Open Theorem 17 (Root B: record-embedding ADM closure)**.** *Derive that changes of slicing, labels and record parametrization are gauge redundancies of the same physical record history. These redundancies must generate first-class constraints:*

$$\boxed{\mathcal{H} \approx 0, \qquad \mathcal{H}_i \approx 0,}$$

with the ADM hypersurface-deformation algebra. If Root B closes, local scalar/vector gravitational sectors are constrained, flow is shift-like in the tested core, only $h_+, h_\times$ propagate, $\lambda = 1/2$ is selected and the two-derivative fixed point is S_{EH}.

Open Theorem 18 (Root B record-action construction bridge)**.** *Root B may audit a projected record action, but it cannot treat that action as a magic input. The medium package must supply the chain*

$$\boxed{\mathcal{D}_{\mathcal{M}} \longrightarrow S_{\mathrm{eff}}^{\mathrm{rec}} \longrightarrow S_{\mathcal{R}}[g_{\mathrm{rec}}, \Psi_{\mathrm{rec}}] + \textit{controlled remainders.}}$$

The object delivered to Root B must be local up to controlled nonlocal tails, have a well-defined two-derivative truncation, match the projected record spectrum $(\mathcal{L}_{\mathcal{M}_0}, \Pi_{\mathcal{R}}, \Pi_{\mathcal{I}})$, give declared tensor or density type to every record variable, remove visible preferred-frame leaks through $\mathcal{G}_{\mathrm{int}}$, and introduce matter through record observables rather than a private lapse, shift, source or metric. If this bridge is missing, Root B remains a target audit rather than a medium-derived ADM closure proof.

Open Theorem 19 (Root B embedding/symplectic closure)**.** *Root B requires showing that the physical object is a relational record history $W_{\mathcal{R}}$, while a foliation/embedding X_F is only a description of that history:*

$$\boxed{X_F \sim X_F', \qquad \Gamma_{\mathrm{phys}} = \Gamma_{\mathcal{R}}/\mathcal{G}_{\mathrm{emb}}.}$$

This equivalence must be a true gauge degeneracy of the record symplectic structure:

$$\boxed{\iota_{\delta_\xi}\Omega_{\mathcal{R}} \approx 0.}$$

Then the embedding generators must take the ADM form

$$\boxed{G[\xi] = D[N^i] + H[N], \qquad D[N^i] = \int d^3x\, N^i \mathcal{H}_i, \qquad H[N] = \int d^3x\, N\mathcal{H},}$$

with

$$\boxed{\mathcal{H}_i \approx 0, \qquad \mathcal{H} \approx 0.}$$

The composition of two embedding changes must be the hypersurface-deformation algebra:

$$\boxed{\{D[N^i], D[M^i]\} = D[[N, M]^i],}$$

$$\boxed{\{H[N], D[M^i]\} = H[\mathcal{L}_M N],}$$

$$\boxed{\{H[N], H[M]\} = D\left[h^{ij}(N\partial_j M - M\partial_j N)\right],}$$

where the structure function h^{ij} is the same inverse spatial metric seen by the record sector. Lapse and shift are then embedding multipliers, not extra physical medium particles:

$$\boxed{S_{\mathrm{loc}} = \int dt\, d^3x \left(\pi^{ij}\dot{h}_{ij} - N\mathcal{H} - N^i\mathcal{H}_i\right), \qquad \dot{N}^2 = \dot{N}_i\dot{N}^i = 0.}$$

This no-kinetic lapse/shift structure is a target output, not a starting assumption. It must follow from the projected record action and its presymplectic form: the momenta conjugate to N and N^i must be primary constraints, and the Dirac algorithm must preserve them without making lapse or shift physical. It may not be imposed merely by choosing an ADM-looking gauge. If this theorem closes, the scalar and vector gravitational sectors are constraints/near-zone bookkeeping, local flow is shift-like, only two tensor modes propagate, the DeWitt value $\lambda = 1/2$ is selected and the two-derivative local fixed point is S_{EH}.

Open Theorem 20 (Root B finite sufficiency checklist)**.** *Root B is reduced to five sufficient conditions on the low-energy record sector:*

$$\boxed{\begin{array}{ll} B1 & \textit{no observable absolute record time,} \\ B2 & \textit{one record geometry } h_{ij}, \\ B3 & \textit{embedding changes are presymplectic null directions,} \\ B4 & \textit{null directions have local generators } D[N^i], H[N], \\ B5 & \textit{generator composition is anomaly-free.} \end{array}}$$

The algebraic danger is a residual anomaly in the normal-normal bracket:

$$\boxed{\{H[N], H[M]\} = D\big[h^{ij}(N\partial_j M - M\partial_j N)\big] + A[N, M].}$$

Root B requires

$$\boxed{A[N, M] = 0}$$

on the local tested record sector. A nonzero A signals an unsuppressed preferred-frame, scalar, vector or non-record tensor structure in the local sector. Equivalently, lapse/shift or local flow has become physical rather than embedding bookkeeping. The action-level check is:

$$\boxed{\dot{N}^2, \quad \dot{N}_i\dot{N}^i, \quad \textit{unsuppressed independent scalar/vector kinetic terms} \quad \textit{are absent.}}$$

Again, this is a derived condition: lapse/shift no-kinetic status must be obtained from the presymplectic/Dirac audit, not inserted as a gauge choice. Thus Root B is not an aesthetic GR assumption. It is the finite test:

$$\boxed{B1 + B2 + B3 + B4 + B5 \Rightarrow \textit{ADM local closure.}}$$

Open Theorem 21 (Root B physical motivation status)**.** *The base-medium picture supports Root B at the following graded level:*

condition	*motivation status*
$B1$	*strong support: time/measurement are operational and relational*
$B2$	*partial: requires Root A one-carrier closure*
$B3$	*open: requires a record presymplectic form* $\Omega_{\mathcal{R}}$
$B4$	*open: requires local moment-map generators* D, H
$B5$	*open: requires anomaly cancellation* $A[N, M] = 0$

A single undivided base medium has no internal ruler, delay or comparison until it self-differentiates into distinguishable record-forming nodes. This strongly supports the absence of an observable absolute record time. It does not by itself prove the canonical statement

$$\boxed{\iota_{\delta_\xi}\Omega_{\mathcal{R}} \approx 0 \quad \Rightarrow \quad G[\xi] = D[N^i] + H[N].}$$

The next Root B step is to construct the low-energy record phase space $\Gamma_{\mathcal{R}}$, its presymplectic form $\Omega_{\mathcal{R}}$, the embedding-variation vector fields δ_ξ and the local generators D, H. The pass condition remains

$$\boxed{\{H[N], H[M]\} - D\big[h^{ij}(N\partial_j M - M\partial_j N)\big] = 0}$$

in the tested local record sector.

Open Theorem 22 (Record phase space and presymplectic Root B construction)**.** *The non-circular Root B construction starts from the covariant record phase space*

$$\boxed{\Gamma_{\mathcal{R}} = \{\Phi_{\mathcal{R}} \text{ solving the record equations}\}/\mathcal{G}_{\text{int}}, \qquad \Phi_{\mathcal{R}} = (g_{\text{rec}}, \Psi_{\text{rec}}),}$$

where $\mathcal{G}_{\text{int}}$ removes internal/non-record redundancies but does not yet quotient embedding freedom. Let the low-energy record action satisfy

$$\boxed{\delta L_{\mathcal{R}} = E_{\mathcal{R}}\delta\Phi_{\mathcal{R}} + d\theta_{\mathcal{R}}(\delta\Phi_{\mathcal{R}}).}$$

Define the covariant presymplectic current

$$\boxed{\omega_{\mathcal{R}}(\delta_1\Phi, \delta_2\Phi) = \delta_1\theta_{\mathcal{R}}(\delta_2\Phi) - \delta_2\theta_{\mathcal{R}}(\delta_1\Phi)}$$

and the slice form

$$\boxed{\Omega_{\mathcal{R},\Sigma} = \int_\Sigma \omega_{\mathcal{R}}.}$$

On shell and without boundary flux, $d\omega_{\mathcal{R}} = 0$, so

$$\boxed{\Omega_{\mathcal{R},\Sigma} = \Omega_{\mathcal{R},\Sigma'}.}$$

This is the formal version of one record history described by different slices. An embedding deformation acts as

$$\boxed{\delta_\xi\Phi_{\mathcal{R}} = \mathcal{L}_\xi\Phi_{\mathcal{R}} \quad \textit{up to internal gauge compensation.}}$$

This statement must be evaluated field by field:

$$\boxed{\begin{array}{rcl} \delta_\xi g_{\text{rec}} &=& \mathcal{L}_\xi g_{\text{rec}}, \\ \delta_\xi\varphi &=& \xi^\mu\partial_\mu\varphi, \\ \delta_\xi T &=& \mathcal{L}_\xi T \quad \textit{for its declared tensor type,} \\ \delta_\xi\rho &=& \mathcal{L}_\xi\rho \quad \textit{including density weight.} \end{array}}$$

If a record variable has no declared tensor or density type, the embedding audit is not yet computable. Root B requires this direction to be presymplectically null:

$$\boxed{\Omega_{\mathcal{R}}(\delta\Phi, \delta_\xi\Phi) \approx 0.}$$

Equivalently, the Noether current

$$\boxed{J_\xi = \theta_{\mathcal{R}}(\mathcal{L}_\xi\Phi_{\mathcal{R}}) - \xi\cdot L_{\mathcal{R}}}$$

must split as

$$\boxed{J_\xi = C_\xi + dQ_\xi, \qquad \int_\Sigma C_\xi = H[N] + D[N^i], \qquad \xi = Nn + N^i e_i.}$$

The pass condition is therefore

$$\boxed{\Omega_{\mathcal{R}}(\delta\Phi, \mathcal{L}_\xi\Phi) = \delta\big(H[N] + D[N^i]\big)}$$

with $H \approx 0$ and $D \approx 0$ in the tested local record sector. Any bulk remainder that is not a constraint or boundary charge is the Root B preferred-frame/scalar/vector anomaly.

The brackets must be computed from $\Omega_{\mathcal{R}}$, not copied from the target algebra. Canonical variables and momenta are read from $\theta_{\mathcal{R}}$; second-class constraints require the Dirac bracket; auxiliary fields must be eliminated before the physical mode count; and boundary terms must be separated before any bulk anomaly $A[N, M]$ is declared.

Open Theorem 23 (Noether split and background-free record-action gate)**.** *The Noether split required by Root B follows if the local tested record action is background-free:*

$$\boxed{L_{\mathcal{R}}^{\text{local}} = L_{\mathcal{R}}[g_{\text{rec}}, \Psi_{\text{rec}}]}$$

up to constraints, gauge choices, boundary terms, integrated-out internal contributions, high-energy suppressed corrections and large-scale-only flow/MOND terms. In that case an embedding deformation is a record diffeomorphism:

$$\boxed{\delta_\xi L_{\mathcal{R}} = \mathcal{L}_\xi L_{\mathcal{R}} = d(\xi \cdot L_{\mathcal{R}}),}$$

and the Noether current can be written as

$$\boxed{J_\xi = C_\xi + dQ_\xi, \qquad C_\xi = N\mathcal{H} + N^i\mathcal{H}_i.}$$

The forbidden local failure is a non-record background structure B:

$$\boxed{L_{\mathcal{R}}^{\text{local}} = L_{\mathcal{R}}[\Phi_{\mathcal{R}}; B].}$$

If B is not a gauge, constraint, boundary, integrated-out, suppressed or large-scale-only object, then the embedding variation leaves a bulk anomaly:

$$\boxed{\mathcal{A}_\xi \sim \frac{\delta L_{\mathcal{R}}}{\delta B}\mathcal{L}_\xi B.}$$

The local-core forbidden B's are:

$$\boxed{T_{\mathcal{M}}, \quad U^\mu_{\mathcal{M}}, \quad v^\mu v^\nu, \quad \textit{species-private metric/clock}, \quad \dot{N}^2, \quad \dot{N}_i\dot{N}^i, \quad \textit{extra record cone.}}$$

Every bulk remainder in J_ξ must therefore be classified as exactly one of:

$$\boxed{\textit{constraint/gauge/boundary/internal/gapped/damped/suppressed/large-scale-only/anomaly.}}$$

Root B requires the final class to vanish in the local tested sector:

$$\boxed{\mathcal{A}_\xi^{\text{local}} = 0 \quad \Longleftrightarrow \quad A[N, M] = 0.}$$

The allowed classes require a structural or scale test. High-energy remainders must scale below the relevant observation row, for example as $(E_{\text{test}}/\Lambda_{\text{UV}})^2$ or $(k_{\text{test}}/\Lambda_{\text{UV}})^2$; local PPN-sensitive terms need at least the corresponding weak-field bound. Large-scale-only terms must activate only outside Solar-System, binary-pulsar and local GW regimes; gapped or damped terms must lack coherent long-range propagation through the tested baseline.

Open Theorem 24 (Background-free two-derivative record-action fixed point)**.** *If Root A supplies one record metric $g_{\rm rec}$, Root B removes all local non-record background structures, and the tested local regime admits a two-derivative effective hierarchy, then the local record action has the form*

$$\boxed{L_{\mathcal{R}}^{\rm local} = \sqrt{-g_{\rm rec}}\,[\alpha R[g_{\rm rec}] - 2\Lambda + L_{\rm m}(g_{\rm rec}, \Psi_{\rm rec})] + dB + O(\partial^4/M_*^2).}$$

At the two-derivative fixed point this is

$$\boxed{S_{\rm EH} + S_{\rm matter}}$$

up to normalization, cosmological term, boundary terms and suppressed higher-derivative corrections. The two-derivative ansatz is itself part of the audit: higher-derivative operators such as R^2, $R_{\mu\nu}R^{\mu\nu}$, Weyl-squared terms or preferred-frame curvature combinations must be suppressed by the hierarchy scale, not used to repair a failed ADM bracket. The leading curvature operator must be the EH operator, with α absorbed into the measured G or identified as a deviation/failure. The forbidden local background leaks are:

$$\boxed{U^\mu_{\mathcal{M}}U^\nu_{\mathcal{M}}\nabla_\mu\Psi\nabla_\nu\Psi, \quad v^\mu v^\nu R_{\mu\nu}, \quad \dot{N}^2, \quad \dot{N}_i\dot{N}^i, \quad g_A^{\mu\nu}\partial_\mu\Psi_A\partial_\nu\Psi_A, \quad C^{\mu\nu}k_\mu k_\nu \not\propto g_{\rm rec}^{\mu\nu}k_\mu k_\nu.}$$

Each such term must be absent, constrained, gauge, boundary-only, integrated-out/internal, high-energy suppressed or large-scale-only. Otherwise the two-derivative fixed point is not GR and Root B fails in the local tested core. Thus the local closure chain is

$$\boxed{\textit{one record metric} + \textit{background-free action} + \textit{two-derivative hierarchy} \Rightarrow S_{\rm EH} + S_{\rm matter}.}$$

Remark (Root B closure status under ORC). Conditional on the Operational Relational Covariance postulate of Section "Primitive Postulates", the Root B open theorem closes as a derived Result. The chain is: (i) ORC's "slicing/labels/parametrization are descriptions, not extra physical structures" forces lapse N and shift N^i into the role of Lagrange multipliers, so $\delta S/\delta N = 0$ and $\delta S/\delta N^i = 0$ yield $\mathcal{H} \approx 0$, $\mathcal{H}_i \approx 0$; (ii) ORC's exclusion of unobservable substrate handles forbids the anomaly $A[N, M]$, so the hypersurface-deformation algebra closes; (iii) by Kuchar–Teitelboim's "geometrodynamics regained" theorem (1976), the unique local two-derivative Hamiltonian with first-class HDA closure and Riemannian configuration space is Einstein–Hilbert plus minimally coupled matter, with DeWitt coefficient $\lambda = 1/2$; (iv) at the linearized level, $h_+, h_\times$ are the only propagating gravitational modes; scalar/vector/breathing sectors are constrained.

Open Theorem 25 (Root C: total-energy source projection)**.** *Derive the far-tail/source theorem:*

$$\boxed{Q_A^{(0)} = \eta E_A/c^2, \qquad S_u = \rho_E,}$$

where E_A includes rest, kinetic, binding and internal/confined energy. No unsuppressed species-private source may remain. If Root C closes, the equivalence-principle source gate and compact internal-energy effacement gate close.

Open Theorem 26 (Root C finite source chain)**.** *Root C reduces to five source conditions:*

$$\boxed{\begin{array}{ll} C1 & S_{\rm m} = S_{\rm m}[g_{\rm rec}, \Psi] \quad \textit{universal matter action}, \\ C2 & N = e^{-u} \quad \textit{or equivalent compression/lapse bridge}, \\ C3 & S_u = (N\sqrt{h})^{-1}\delta S_{\rm m}/\delta u = \rho_E, \\ C4 & \nabla^2 u = -\kappa\rho_E \quad \textit{far-tail Gauss law}, \\ C5 & \sigma_{\rm species}^{\rm long,local} = 0. \end{array}}$$

Then an isolated source obeys

$$\boxed{\oint_{\partial B_R} \nabla u \cdot dS = -\kappa \int_{B_R} \rho_E \, d^3x, \qquad u(r) = \frac{Q^{(0)}}{r} + O(r^{-2}),}$$

and therefore

$$\boxed{Q^{(0)} = \eta \, E_{\text{total}}/c^2.}$$

The substantive content of Root C is not the far $1/r$ integral but the requirement that the compression/lapse source equals the total physical energy for all stable record matter:

$$\boxed{S_u = \rho_E}$$

with no unsuppressed species-private remainder:

$$\boxed{S_u \neq \rho_E + \sigma_{\text{species}} \quad \textit{for physical long-range local } \sigma_{\text{species}}.}$$

Species-specific structure is allowed only if it is an internal divergence, gauge, boundary multipole, high-energy suppressed, short-range/confined or already included in ρ_E. Internal, binding, gluon-like and confined energy must contribute to E_{total} while not producing an independent external cone:

$$\boxed{E_{\text{total}} = E_{\text{rest}} + E_{\text{kinetic}} + E_{\text{binding}} + E_{\text{internal}}.}$$

Here binding includes self-gravitational binding in the strong-equivalence audit. Self-gravitating bodies must not fall differently merely because a different fraction of their total mass is stored as gravitational binding energy. In local tests this is the Nordtvedt-type failure mode; in compact binaries it appears as a sensitivity/dipole-radiation audit.

Open Theorem 27 (C3 lapse-variation source theorem)**.** *Assume the stable record matter sector satisfies*

$$\boxed{S_{\text{m}} = S_{\text{m}}[g_{\text{rec}}, \Psi], \qquad N = e^{-u},}$$

and has no unsuppressed explicit species-private u-coupling outside the record metric. In canonical form,

$$\boxed{S_{\text{m}} = \int dt \, d^3x \, \left(\pi_A \dot{\psi}_A - N \mathcal{H}_{\text{m}} - N^i \mathcal{H}_{i,\text{m}} \right).}$$

At fixed $h_{ij}, N^i, \psi_A, \pi_A$,

$$\boxed{\delta N = -N \delta u, \qquad \delta_u S_{\text{m}} = \int dt \, d^3x \, N \mathcal{H}_{\text{m}} \delta u.}$$

Therefore

$$\boxed{S_u = \frac{1}{N\sqrt{h}} \frac{\delta S_{\text{m}}}{\delta u} = \frac{\mathcal{H}_{\text{m}}}{\sqrt{h}} = \rho_E.}$$

This is the Root C bridge: changing compression/lapse changes the local process rate, and the generator of local process-time translation is the Hamiltonian density. Internal, binding, gluon-like and confined energy are included because

$$\boxed{\mathcal{H}_{\text{m}} = \mathcal{H}_{\text{rest}} + \mathcal{H}_{\text{kinetic}} + \mathcal{H}_{\text{binding}} + \mathcal{H}_{\text{internal}}.}$$

Root C fails only if a physical long-range local remainder survives:

$$\boxed{S_u = \rho_E + \sigma^{\text{long,local}}_{\text{species}}, \qquad \sigma^{\text{long,local}}_{\text{species}} \neq 0.}$$

Thus the C3 gate is: all stable record matter must see u only through the same record metric/lapse, except for gauge, constrained, boundary, internal, short-range or suppressed terms already accounted for in ρ_E.

Open Theorem 28 (Metric-exhaustion and no species-private u-coupling)**.** *If the stable local matter sector is exhausted by one record metric,*

$$\boxed{S_{\text{m}} = S_{\text{m}}[g_{\text{rec}}, \Psi],}$$

then all allowed local u-dependence is metric/lapse dependence. Species labels may appear in masses, charges, spin, internal Hamiltonians, confined sectors, kinetic matrices, multipoles and finite-size corrections, but not as a separate long-range local term

$$\boxed{\Delta S_A = \int d^4x\, N\sqrt{h}\, F_A(u, \Psi_A).}$$

If such a term is physical and unsuppressed, then

$$\boxed{S_u = \rho_E + \sigma_A,}$$

so u becomes a species-readable scalar background rather than just the record lapse/compression variable. This is a fifth-force/source anomaly: two systems with the same total energy but different composition would source or fall differently. Therefore the local Root C gate is

$$\boxed{F^{\text{long,local}}_A = 0 \quad \Longleftrightarrow \quad \sigma^{\text{long,local}}_{\text{species}} = 0,}$$

unless the term is already included in ρ_E, or is gauge, constrained, boundary, internal, short-range/confined or high-energy suppressed. In the common pressure projection language,

$$\boxed{\Pi_{\text{grav}}(T_A) = q_A \mathcal{G}, \qquad q_A = \eta E_A/c^2}$$

must hold at monopole order; species-specific tail data may survive only as spin, charge, color, phase, multipole, internal, short-range or suppressed structure.

Remark (Root C closure status under ORC). Conditional on the Operational Relational Covariance postulate, the Root C open theorems close as derived Results once Root A and Root B are in place. The argument is: (i) Root A under ORC yields a single record metric $g^{\mu\nu}_{\text{rec}}$; Root B under ORC yields $S = S_{\text{EH}}[g_{\text{rec}}] + S_{\text{matter}}[g_{\text{rec}}, \Psi]$; (ii) the variational stress-energy tensor $T^{\mu\nu} = (2/\sqrt{-g})\, \delta S_{\text{matter}}/\delta g_{\mu\nu}$ is the unique gravitational source compatible with the diffeomorphism invariance of S_{matter} and contains rest, kinetic, binding, internal/confined and field energies of every stable matter species; (iii) ORC's "one operational geometry for every observer" clause forbids species-private cones and species-private u-coupling, so $C5$ ($\sigma^{\text{long,local}}_{\text{species}} = 0$) is automatic; (iv) $\nabla^2 u = -\kappa\rho_E$ and $Q^{(0)} = \eta E_{\text{total}}/c^2$ then follow from the linearized Einstein equation in the weak static limit. Residual quantitative items — the numerical value of G, the smallness of Λ, and Standard-Model species-by-species verification of the universal coupling — remain open as part of the broader G-closure and matter-emergence programs.

Open Theorem 29 (Local gravity core closure)**.** *If Root A, Root B and Root C are derived from $\mathcal{M}$, then the local RMG gravity core is closed:*

$$\boxed{\textit{one operational spacetime} \quad + \quad S_{\mathrm{EH}} + S_{\mathrm{matter}} \quad + \quad \textit{universal total-energy source.}}$$

The remaining sectors – strong-field compact interiors, MOND reentry, cosmology and detailed matter emergence – then become extension windows, not loose foundations of the local gravitational core.

Remark (full gravity-core closure under ORC). Conditional on the Operational Relational Covariance postulate, the three roots close as derived Results (remarks above on Root A, Root B, Root C), and therefore so does this "Local gravity core closure" openthm. The unified statement is: under ORC plus the minimal RMG structural commitments (one base medium $\mathcal{M}$ with stable phase $\mathcal{M}_0$; locality and two-derivative low-energy hierarchy), the low-energy operational sector is exactly

$$\boxed{S_{\mathrm{low}} = \frac{c^4}{16\pi G}\int d^4x\,\sqrt{-g}\,(R-2\Lambda) + S_{\mathrm{matter}}[g,\Psi] + \text{suppressed corrections},}$$

with one record cone, two TT graviton modes at $c_{\mathrm{GW}} = c$, no scalar or vector gravity waves, no preferred frame, universal free fall, and gravitational source equal to the total physical energy-momentum. This is the tested local GR core.

Status note. The ORC postulate is the single controlling axiom of this closure; its microscopic derivation from a deeper relational substrate remains the central long-term research program. Candidate routes include emergent gauge redundancy from a UV principle, ORC as an IR fixed point of a nonlinear relational substrate, and consistency arguments showing that ORC-violating theories produce internal pathologies. No such derivation has yet been completed. Until then, the gravitational core is closed conditional on ORC.

Open Theorem 30 (Non-circular local-core proof order)**.** *The local gravity proof must close in the following order:*

$$\boxed{\mathcal{M}_{\mathrm{stable}} \Rightarrow \textit{Root A} \Rightarrow g_{\mathrm{rec}} \Rightarrow \textit{Root B} \Rightarrow S_{\mathrm{EH}} + S_{\mathrm{matter}} \Rightarrow \textit{Root C} \Rightarrow S_u = \rho_E.}$$

Equivalently:

$$\boxed{\begin{array}{lcl} \textit{Root A} & : & \textit{one common external record carrier/metric}, \\ \textit{Root B} & : & \textit{record re-embedding gauge} \;\Rightarrow\; \textit{ADM/EH}, \\ \textit{Root C} & : & \textit{metric/lapse matter coupling} \;\Rightarrow\; \textit{total-energy source.} \end{array}}$$

The proof must obey the no-circularity rules:

$$\boxed{\begin{array}{l} \textit{do not assume ADM to prove Root A}, \\ \textit{do not assume Einstein equations to prove Root B}, \\ \textit{do not assume the equivalence principle to prove Root C}, \\ \textit{do not use MOND/cosmology to compensate for a local-core failure.} \end{array}}$$

The microscopic object still needed from the base medium is an effective record action

$$\boxed{S^{\mathrm{rec}}_{\mathrm{eff}}[\mathcal{M}_0]}$$

with one record metric, no local background leak, a covariant presymplectic form, universal matter metric coupling and no explicit species-private u-coupling. This is the next exact derivation target.

12 Leading-Order Relational Base Derivation

The current internal result is a leading-order derivation of the tested local gravity core from a relational base-medium action candidate. It is not the final microscopic theory: the exact coefficients, matter spectrum, MOND scale and strong-field branch remain to be computed.

Result 7 (Relational base action candidate)**.** *The base-medium is represented without external rods, clocks or background space by relational variables*

$$\boxed{R_{ab}, \qquad \vartheta_{ab}, \qquad K_{ab}.}$$

Here R_{ab} denotes adjacency/separation records, ϑ_{ab} the common rhythm/phase records, and K_{ab} internal coupling/shape records. The candidate base action has Jacobi/constraint form

$$\boxed{S_{\mathcal{M}}^{\mathrm{rel}} = \int d\lambda \left[\Theta_{\mathcal{M}}(R, \vartheta, K; \dot{R}, \dot{\vartheta}, \dot{K}) - \alpha C_0 - \beta^a C_a - \gamma C_\nu\right].}$$

λ is only a bookkeeping parameter, not physical time. The constraints have the intended meanings

$$\boxed{C_0 = \textit{common process constraint}, \qquad C_a = \textit{relabel/re-embedding constraint}, \qquad C_\nu = \textit{one-rhythm/harmonic sele}}$$

Thus the base does not begin with spacetime. Spacetime variables are allowed only as coarse-grained record variables.

Result 8 (Continuum core route)**.** *Under smooth coarse-graining, local isotropy, one common record carrier and no physical background-label leak, the continuum record variables are*

$$\boxed{h_{ij} = h_{ij}[R, \vartheta, K], \qquad \pi^{ij} = \pi^{ij}[R, \vartheta, K].}$$

Relabel/re-embedding redundancy gives the momentum constraint:

$$\boxed{C_a \longrightarrow H_i.}$$

Small relational loop defects give curvature:

$$\boxed{\Omega_{\mathrm{loop}} - 1 \longrightarrow R^i{}_{jkl}(h), \qquad V_{\mathrm{loop}} \longrightarrow \int d^3x \sqrt{h}\,(R(h) - 2\Lambda).}$$

The relational kinetic term decomposes into physical shear plus constrained volume/gauge directions. In the protected continuum limit this gives the DeWitt kinetic form

$$\boxed{T_{\mathrm{rec}} \longrightarrow \frac{1}{\sqrt{h}}\left(\pi_{ij}\pi^{ij} - \frac{1}{2}\pi^2\right).}$$

Therefore the leading-order local record Hamiltonian is

$$\boxed{H = \frac{1}{\sqrt{h}}\left(\pi_{ij}\pi^{ij} - \frac{1}{2}\pi^2\right) - \sqrt{h}\,(R(h) - 2\Lambda) + H_{\mathrm{matter}}.}$$

Together with H_i, this is the ADM/Einstein-Hilbert local fixed point at leading order.

Result 9 (Radiation and source consequence)**.** *The metric perturbation splits operationally as*

$$\boxed{\delta h_{ij} = s_{ij} + \frac{1}{3} h_{ij} b + \textit{gauge/constraint pieces.}}$$

The common breathing variable b is not a separate free radiation field in the tested local sector; it is hidden by common-mode self-calibration and by the Hamiltonian constraint. The observable radiation is the shear sector

$$\boxed{s_{ij} = h_{ij}^{TT},}$$

which carries two tensor modes. Matter couples to the same record metric, so the local source is total energy rather than a species-private tail charge.

Gate 3 (Leading-order status)**.** *RMG currently has a conditional leading-order route*

$$\boxed{\mathcal{M}_{\rm rel} \Rightarrow \textit{record geometry} \Rightarrow \textit{ADM/EH local core} \Rightarrow \textit{two tensor GW modes} \Rightarrow \textit{total-energy source.}}$$

This is the present leading-order core result. It becomes a full derivation only after the exact coarse-graining map, the constants G, Λ, the stable spectrum, the matter couplings, the MOND reentry scale and the strong-field branch are computed from the base-medium rather than inserted as targets.

Open Theorem 31 ($S_{\rm eff}^{\rm rec}$ derivation pipeline)**.** *Start from a base-medium action/evolution functional*

$$\boxed{S_{\mathcal{M}}[\Phi_{\mathcal{M}}], \qquad \Phi_{\mathcal{M}} = \Phi_0 + \delta\Phi,}$$

around a stable phase $\mathcal{M}_0$. Linearization gives

$$\boxed{\delta^2 S_{\mathcal{M}} = \frac{1}{2}\langle \delta\Phi, \mathcal{L}_{\mathcal{M}_0}\delta\Phi\rangle.}$$

The fluctuation space must split into record and internal sectors:

$$\boxed{\delta\Phi = R + I, \qquad R = \Pi_{\mathcal{R}}\delta\Phi, \qquad I = \Pi_{\mathcal{I}}\delta\Phi,}$$

with

$$\boxed{\Pi_{\mathcal{R}} + \Pi_{\mathcal{I}} = 1, \qquad \Pi_{\mathcal{R}}\Pi_{\mathcal{I}} = 0.}$$

The record effective action is then the low-energy functional

$$\boxed{e^{iS_{\rm eff}^{\rm rec}[R]} = \int \mathcal{D}I\, e^{iS_{\mathcal{M}}[R,I]}.}$$

Equivalently,

$$\boxed{S_{\rm eff}^{\rm rec}[R] = -i \log \int \mathcal{D}I\, e^{iS_{\mathcal{M}}[R,I]}.}$$

The record-sector principal kernel must factorize:

$$\boxed{S_{\mathcal{R}}^{(2)} \sim \int R_A\, Q_{AB}^{\mu\nu}\partial_\mu\partial_\nu R_B, \qquad Q_{AB}^{\mu\nu} = G^{\mu\nu}K_{AB},}$$

so that one record metric is defined by $g^{\mu\nu}_{\rm rec} \sim G^{\mu\nu}$. The internal integration may renormalize constants, add binding energy, multipoles, finite-size terms, high-energy suppressed operators or large-scale-only corrections. It must not leave local unsuppressed

$$\boxed{U^\mu_{\mathcal{M}}, \quad v^\mu v^\nu, \quad \text{species-private metric}, \quad \text{explicit species-private } u\text{-coupling}, \quad \dot{N}^2, \quad \dot{N}_i \dot{N}^i, \quad \text{extra record cone.}}$$

If this pipeline succeeds, the same microscopic object supplies the raw material for Root A, Root B and Root C. If it fails, the failure is localized as either non-factorization, background leak, or source anomaly.

Open Theorem 32 (Minimal base-medium input package)**.** *To compute $S^{\rm rec}_{\rm eff}$, the base-medium model must provide the input package*

$$\boxed{\mathcal{D}_{\mathcal{M}} = (\Phi_{\mathcal{M}}, S_{\mathcal{M}}, \mathcal{M}_0, \ell_*, \Omega_{\mathcal{M}}, \mathcal{G}_{\rm int}, \mathcal{O}_{\rm rec})\,.}$$

Here $\Phi_{\mathcal{M}}$ denotes the microscopic/collective medium variables, $S_{\mathcal{M}}$ the action or evolution functional, $\mathcal{M}_0$ the stable low-energy phase, ℓ_* the minimal operational node scale, $\Omega_{\mathcal{M}}$ the symplectic or presymplectic structure, $\mathcal{G}_{\rm int}$ the internal redundancy group, and $\mathcal{O}_{\rm rec}$ the operational record observables. Around $\mathcal{M}_0$,

$$\boxed{\Phi_{\mathcal{M}} = \Phi_0 + \delta\Phi, \qquad \delta^2 S_{\mathcal{M}} = \frac{1}{2}\langle \delta\Phi, \mathcal{L}_{\mathcal{M}_0}\delta\Phi\rangle.}$$

The spectral target is

$$\boxed{\mathrm{Spec}(\mathcal{L}_{\mathcal{M}_0}) = \mathcal{B}_{\rm rec} \cup \mathcal{B}_{\rm int}, \qquad N_{\rm cone}(\mathcal{B}^{\rm gapless}_{\rm rec}) = 1}$$

as one external record carrier/cone, not one scalar field. The projectors must be built from observability and spectrum, not by assuming GR:

$$\boxed{\Pi_{\mathcal{R}} + \Pi_{\mathcal{I}} = 1, \qquad \Pi_{\mathcal{R}}\Pi_{\mathcal{I}} = 0.}$$

At quadratic order, internal integration has the block form

$$\boxed{S^{(2)} = \frac{1}{2}\begin{pmatrix} R & I \end{pmatrix}\begin{pmatrix} L_{RR} & L_{RI} \\ L_{IR} & L_{II} \end{pmatrix}\begin{pmatrix} R \\ I \end{pmatrix}, \qquad L_{\rm eff} = L_{RR} - L_{RI}L_{II}^{-1}L_{IR}.}$$

The Schur-complement correction may renormalize the record kernel, binding energy, multipoles or suppressed operators. It must not generate an extra cone, local preferred-frame tensor, species-private metric, explicit species-private u-coupling or lapse/shift kinetic terms. Without this input package, RMG-1 has a well-defined proof target; the microscopic derivation remains open.

Status of the package. *At this stage $\mathcal{D}_{\mathcal{M}}$ is a required framework, not yet a completed microscopic specification. Before Root A can be calculated, it must be promoted to at least one explicit ansatz whose quadratic operator, projectors and record/internal split can actually be computed.*

Additional input requirements. *The node scale ℓ_* must be assigned a clear status. In a first toy model it may be treated as a primitive coarse-graining input; in the final theory it should be derived or calibrated from $S_{\mathcal{M}}$ and stable record dynamics. The form $\Omega_{\mathcal{M}}$ must identify physical, gauge, constrained and presymplectic-null directions; the spectrum of $\mathcal{L}_{\mathcal{M}_0}$ alone is not enough, because a Hessian mode may be gauge or constrained rather than physical.*

The record-internal coupling L_{RI} is a central physics input. Its symmetry, sign, scale and derivative order must be specified. The correction $-L_{RI}L_{II}^{-1}L_{IR}$ may renormalize record dynamics, but it must not change the principal cone, create a long-range private source, or flip the sign of a physical kinetic term.

Open Theorem 33 (Candidate base-medium order-parameter ansatz)**.** *A first calculable microscopic target is a single base-medium order parameter, not many independent fundamental substances:*

$$\boxed{\Phi_{\mathcal{M}} = \Psi_{\mathcal{M}} = \sqrt{n}\, e^{i\theta}\chi.}$$

Here n is local node-density/pressure amplitude, θ is phase or medium rhythm, and χ denotes internal shape/orientation data of the same medium. This ansatz is allowed only if χ does not become an independent long-range spacetime carrier. It must be internal, confined, gauge, constrained, suppressed, or part of record matter sharing the same carrier. Before Root A is calculated, χ must be assigned a definite mathematical type: scalar, vector, tensor, group-valued field, constrained composite variable, or topological label. Merely saying "internal shape" is not yet a microscopic specification.

A minimal positive-energy candidate functional is

$$\boxed{E_{\mathcal{M}} = \int d^3X \left[\frac{\kappa_n}{2}(\nabla n)^2 + \frac{\kappa_\theta}{2} n(\nabla\theta)^2 + V(n) + E_{\mathrm{int}}(n, \theta, \chi, \nabla\chi)\right],}$$

with a stable phase

$$\boxed{V'(n_0) = 0, \qquad V''(n_0) > 0.}$$

Schematically,

$$\boxed{S_{\mathcal{M}} = \int d\tau \left[\Theta_{\mathcal{M}}(\Phi_{\mathcal{M}}, \dot{\Phi}_{\mathcal{M}}) - E_{\mathcal{M}}[\Phi_{\mathcal{M}}]\right],}$$

where $\Theta_{\mathcal{M}}$ defines the symplectic structure. The stable phase expansion is

$$\boxed{n = n_0 + \delta n, \qquad \theta = \nu_0\tau + \delta\theta, \qquad \chi = \chi_0 + \delta\chi.}$$

The Root A spectral target is still

$$\boxed{N_{\mathrm{cone}}(\mathcal{B}_{\mathrm{rec}}^{\mathrm{gapless}}) = 1.}$$

This means one observable external carrier/cone, not one scalar excitation. The bookkeeping parameter τ, any preferred flow, and any extra observable carrier branch must not survive as local observables in $S_{\mathrm{eff}}^{\mathrm{rec}}$. The background form $\theta = \nu_0\tau + \delta\theta$ is allowed only if τ is protected from becoming an observable absolute time. The required condition is

$$\boxed{\delta_\tau \mathcal{O}_{\mathrm{rec}} \approx 0}$$

for all physical record observables, up to gauge or constraint terms. The allowed routes are a parametrized action with a reparametrization constraint, a relational clock gauge fixing, or a presymplectic null direction generated by changes of the time label.

The local Lorentz behavior of the record sector must also be explicit. Either it emerges from factorization,

$$\boxed{Q_{AB}^{\mu\nu} = G^{\mu\nu}K_{AB},}$$

with one common principal cone, or a covariant low-energy record action is stated as a target fixed point. It is not enough to assume observable Lorentz invariance while leaving a physical medium time, medium velocity or species-private cone in the local record sector. The long compression variable u should arise from a pressure/density deficit, for example $u \sim -\delta \ln n$, and its weak static equation must be constraint-like:

$$\boxed{\nabla^2 u = -\kappa\rho_E + \textit{allowed corrections},}$$

so that u remains a constraint variable rather than an independent propagating scalar mode. This ansatz is a test model, not a proof: if it leaves a physical medium time, preferred frame or extra cone, it must be constrained or replaced.

Open Theorem 34 (Working toy instantiation of the input package)**.** *The first calculable test bench is allowed to be wrong; its purpose is to make $\mathcal{D}_{\mathcal{M}}$ concrete enough that the Root A spectrum and the spin-2 gate can be tested. Take*

$$\boxed{\Phi^{\mathrm{toy}}_{\mathcal{M}} = (\phi, \theta, Q_{ij}), \qquad \phi = \sqrt{n},}$$

where Q_{ij} is a real symmetric traceless internal strain tensor,

$$\boxed{Q_{ij} = Q_{ji}, \qquad Q^{i}{}_{i} = 0.}$$

The toy stable phase is isotropic:

$$\boxed{\phi = v, \qquad \theta = \nu_0 \tau, \qquad Q_{ij} = 0.}$$

Tensor information, if any, must therefore come from fluctuations of Q_{ij} and their record-visible transverse-traceless projection, not from a preferred background direction.

A minimal schematic toy energy is

$$\boxed{E_{\mathrm{toy}} = \int d^3X \left[\frac{k_\phi}{2}(\nabla\phi)^2 + \frac{k_\theta}{2}\phi^2(\nabla\theta)^2 + \frac{\lambda_\phi}{4}(\phi^2 - v^2)^2 + \frac{k_Q}{2}(\nabla Q_{ij})^2 + \frac{\mu_Q^2}{2}Q_{ij}Q_{ij} + \frac{g_Q}{2}\phi^2 Q_{ij}Q_{ij} + E_{\mathrm{mix}}\right]}.$$

The action must be parametrized or presymplectic, for example

$$\boxed{S_{\mathrm{toy}} = \int d\tau \left[\Theta_{\mathrm{toy}} - \alpha C_0 - \beta^a C_a - E_{\mathrm{toy}}\right],}$$

so that the bookkeeping label τ and relabeling variables do not become physical background structures.

Expanding

$$\boxed{\phi = v + \eta, \qquad \theta = \nu_0\tau + \vartheta, \qquad Q_{ij} = q_{ij},}$$

the quadratic target has the schematic form

$$\boxed{\delta^2 E_{\mathrm{toy}} = \frac{1}{2}\eta(-k_\phi\nabla^2 + m_\eta^2)\eta + \frac{1}{2}\vartheta(-k_\theta v^2\nabla^2)\vartheta + \frac{1}{2}q_{ij}(-k_Q\nabla^2 + m_Q^2)q_{ij} + L^{(2)}_{\mathrm{mix}},}$$

with

$$\boxed{m_\eta^2 = 2\lambda_\phi v^2, \qquad m_Q^2 = \mu_Q^2 + g_Q v^2.}$$

The conservative spectrum reading is: η is gapped if $m_\eta^2 > 0$; ϑ is the candidate common phase/cone branch but not the graviton; q_{ij} is gapped if $m_Q^2 > 0$; and $L^{(2)}_{\mathrm{mix}}$ must not change the principal cone. For the conservative Root A toy, $m_Q^2 > 0$ is required unless the Q-sector is explicitly being tested as the shared-cone TT candidate. An unintended massless Q_{ij} branch carries five symmetric-traceless components and can create extra long-range branches, threatening $N_{\mathrm{cone}} = 1$. This version may help Root A while still failing to produce tensor gravity.

Allowed toy mixing terms must be derivative-order and symmetry controlled, for example

$$\boxed{a_1\eta(\nabla\vartheta)^2, \quad a_2\eta Q_{ij}Q_{ij}, \quad a_3 Q_{ij}\partial_i\vartheta\,\partial_j\vartheta, \quad a_4\eta\,\partial_k Q_{ij}\partial_k Q_{ij}.}$$

They are allowed only if, after expansion and internal integration, they do not split the record cone, introduce a preferred tensor or create a long-range species-private source.

To test the tensor gate, decompose

$$\boxed{q_{ij} = q_{ij}^{TT} + \partial_i V_j + \partial_j V_i + \left(\partial_i\partial_j - \frac{1}{3}\delta_{ij}\nabla^2\right) B + \frac{1}{3}\delta_{ij} H.}$$

The tensor candidate passes only if

$$\boxed{\operatorname{rank} K_{Q-\mathrm{strain},TT} = 2, \qquad \textit{positive kinetic sign}, \qquad c_{TT} = c_{\mathrm{rec}},}$$

and all scalar/vector pieces are gauge, constrained, near-zone or suppressed. Rank zero means no tensor gravity; rank greater than two means extra local shear/scalar/vector radiation; a wrong sign, mass term or different cone also fails. Thus the toy model localizes failure instead of hiding it: extra cones, ghosts, preferred frames or species-private sources identify which part of $\mathcal{D}_{\mathcal{M}}$ must be changed.

Result 10 (Stable-record coherent amplitude completion)**.** *The ansatz $\Phi_{\mathcal{M}} = \sqrt{n}e^{i\theta}\chi$ by itself is not yet a microscopic proof of the square law. The square law is derived if the projected long-range stable record sector has the coherent endpoint-counting completion*

$$\boxed{\rho := |\Phi_{\mathcal{M}}| = \sqrt{n}}$$

and the first-order record action

$$\boxed{S_{\mathrm{rec}} = \int d\lambda\, d^3q \left[P_A \dot{R}_A - \alpha\, \mathcal{C}_{\mathrm{rec}}\right],}$$

with

$$\boxed{\mathcal{C}_{\mathrm{rec}} = \frac{1}{2}\left[K^{AB}P_A P_B + K_{AB}\left(c^2\rho^2\nabla_{\mathrm{rec}}R_A\nabla_{\mathrm{rec}}R_B + m_A^2\rho R_A R_B\right)\right].}$$

The multiplier α is the common process multiplier, not an external time. In local common-rhythm proper gauge, $d\tau = \alpha d\lambda$, the momenta eliminate to

$$\boxed{S^{(2)}_{\mathrm{rec}} = \frac{1}{2}\int d\tau\, d^3q\, K_{AB}\left[\partial_\tau R_A \partial_\tau R_B - c^2\rho^2\nabla_{\mathrm{rec}}R_A\nabla_{\mathrm{rec}}R_B - m_A^2\rho R_A R_B\right].}$$

Therefore

$$\boxed{C_t = 1, \qquad C_0 = \rho, \qquad C_x = \rho^2.}$$

The local dispersion relation is

$$\boxed{\omega^2 = c^2\rho^2 k^2 + m_A^2\rho.}$$

Consequently, with $s = \rho^{1/2} = n^{1/4}$,

$$\boxed{\omega_0 \sim s, \qquad \ell_{\mathrm{coord}} \sim s, \qquad c_{\mathrm{coord}} = cs^2.}$$

This is the stable-record completion underlying the reciprocal clock/rod response. The order parameter alone does not yield the result; the endpoint-counting record completion is the additional calculable structure.

Result 11 (Endpoint-counting exponent selection)**.** *The powers in the coherent completion have a relational node-link origin. A local gap term has one active endpoint; hence its leading coherent amplitude weight is*

$$\boxed{L_a \propto \rho_a.}$$

A spatial link has two active endpoints and must vanish if either endpoint is inactive. Endpoint symmetry and minimal coherent order then give

$$\boxed{J_{ab} \propto \rho_a \rho_b.}$$

In a smooth patch, $\rho_a, \rho_b \to \rho$, so the stable record sector has

$$\boxed{\text{gap} \sim \rho, \qquad \text{link stiffness} \sim \rho^2.}$$

This selects $C_t : C_0 : C_x = 1 : \rho : \rho^2$, and therefore the square law $c_{\rm coord}/c = s^2$ with $s = \rho^{1/2}$.

Open Theorem 35 (Candidate stable-phase spectrum audit)**.** *For the candidate*

$$\boxed{\Phi_{\mathcal{M}} = \sqrt{n}\, e^{i\theta} \chi,}$$

write

$$\boxed{n = n_0 + \eta, \qquad \theta = \nu_0 \tau + \vartheta, \qquad \chi = \chi_0 + \xi.}$$

The quadratic spectrum should split schematically as

$$\boxed{\delta^2 S_{\mathcal{M}} = S^{(2)}_{\eta\eta} + S^{(2)}_{\vartheta\vartheta} + S^{(2)}_{\xi\xi} + S^{(2)}_{\rm mix}.}$$

The desired Root A pattern is:

sector	*required low-energy status*
η	*gapped, constrained, near-zone or suppressed*
ϑ	*common rhythm/cone variable, not by itself the graviton*
ξ	*internal/confined/gauge/gapped/suppressed or shared-cone shear/matter*
$S_{\rm mix}$	*renormalization only; no extra cone*

The phase branch may have

$$\boxed{\omega^2 = c_{\rm rec}^2 k^2 + O(k^4), \qquad G^{\mu\nu} k_\mu k_\nu = 0,}$$

and can serve as the candidate common record cone. But this is only Root A input. It does not by itself prove gravitational dynamics:

$$\boxed{\text{one cone} \neq \text{two tensor gravitons.}}$$

Identifying the phase/acoustic branch with the graviton would reduce the framework to scalar acoustic gravity, which is excluded by the two-tensor-mode requirement. The local-GR route is

$$\boxed{\text{one common record cone} \xrightarrow{\text{Root B}} \text{ADM constraints} \xrightarrow{\text{two-derivative fixed point}} S_{\rm EH} \Rightarrow \text{two tensor gravitational modes.}}$$

This candidate passes the first spectrum check only if the amplitude mode is not a long-range scalar gravity mode, internal modes do not generate extra cones, the acoustic cone is universal for record matter, and the tensor sector is derived by Root B rather than assumed.

Open Theorem 36 (Composite record metric and spin-2 emergence gate)**.** *The record metric must be a composite response of the base medium:*

$$\boxed{g^{\mu\nu}_{\rm rec} = \mathcal{G}^{\mu\nu}[\Phi_{\mathcal{M}}], \qquad g^{\mu\nu}_{\rm rec} = \bar{g}^{\mu\nu} + \delta g^{\mu\nu}_{\rm rec}.}$$

The phase/acoustic branch may define the common cone, but the gravitational field is the low-energy dynamics of the composite record geometry, not simply the scalar phase ϑ:

$$\boxed{\delta g^{\mu\nu}_{\rm rec} = \frac{\delta \mathcal{G}^{\mu\nu}}{\delta \Phi_{\mathcal{M}}} \delta \Phi_{\mathcal{M}} + \cdots .}$$

In spatial language,

$$\boxed{\delta h_{ij} = h^{TT}_{ij} + \partial_i V_j + \partial_j V_i + \left(\partial_i \partial_j - \frac{1}{3}\delta_{ij}\nabla^2\right) B + \frac{1}{3}\delta_{ij} H.}$$

Root B must make the scalar and vector pieces constraint, gauge or near-zone, leaving only

$$\boxed{h^{TT}_{ij}}$$

as propagating vacuum gravitational data:

$$\boxed{\#(\text{propagating vacuum gravitational DOF}) = 2.}$$

Possible microscopic sources of this tensor/shear sector are anisotropic stress/strain of χ, collective deformation of the record propagation kernel $G^{\mu\nu}$, topological/oscillon-network elasticity, or emergent embedding geometry after quotienting record histories. None may create a second long-range matter cone or private spacetime. Thus:

$$\boxed{\textit{acoustic cone} = \textit{kinematics}, \qquad \textit{ADM closure} = \textit{gravitational dynamics.}}$$

If the candidate supplies only a scalar acoustic mode, it fails the spin-2 gate.

Gate 4 (Tensor-closure reduction)**.** *The spin-2 issue is not closed in RMG-1. It is reduced to a finite calculation on the candidate record action. The candidate may pass only if the record-visible χ-strain has a transverse-traceless projection and its quadratic kinetic matrix has exactly two positive, luminal, common-cone radiative modes. Equivalently:*

$$\boxed{\textit{candidate tensor closure} \iff \operatorname{rank} K_{\chi-\rm strain} = 2 \ \textit{on the physical record sector.}}$$

Rank zero means no tensor gravity; rank greater than two means extra local shear/scalar/vector radiation; a wrong sign, mass term, or different cone also fails. The detailed proof attempt and projector algebra belong in the work dossier, not in the core file, until the kinetic-rank test is actually derived from $S_{\mathcal{M}}$ or $S^{\rm rec}_{\rm eff}$.

This theorem remains open in RMG-1.

13 Allowed Deviation Windows

Additional sectors may be introduced only if they do not modify the local GR core.

Allowed windows:

$$\text{MOND-like low-acceleration response,}$$

$$\text{cosmological background state of the medium,}$$

$$\text{matter-sector emergence,}$$

$$\text{high-energy corrections.}$$

Rule:

$$\boxed{\text{deviations must not violate the local universal metric, } \gamma = \beta = 1, \text{ or } c_{\mathrm{GW}} = c.}$$

They also must not introduce an independent local $u^\mu_{\mathcal{M}}$ into the tested core.

Gate 5 (Flow/vorticity split). *Medium flow and vorticity are allowed only in two protected roles. In the tested local regime they must be hidden inside ADM constraint variables or near-zone fields:*

$$\boxed{\mathcal{A}_i,\ u \quad \textit{constraint/near-zone only, not independent vacuum radiation.}}$$

In large-scale or low-acceleration regimes they may re-enter as an effective medium correction:

$$\boxed{\mathbf{a} = \mathbf{a}_{\mathrm{GR}} + \mathbf{a}_{\mathrm{med}}.}$$

The correction must satisfy:

$$\boxed{\mathbf{a}_{\mathrm{med}} \to 0 \quad \textit{in local tested high-acceleration regimes,}}$$

while it may become nonzero in galactic low-acceleration or cosmological background regimes.

Gate 6 (Local flow decoupling). *In the tested local core, medium flow and vorticity may enter only as constraint, shift, near-zone or lower-order transport data. They must not modify the physical record-sector principal tensor:*

$$\boxed{\Delta P_A^{\mathrm{local}}(k) \not\supset \alpha_A v^\mu v^\nu k_\mu k_\nu \quad \textit{for external record-forming } A.}$$

Equivalently, the preferred-frame PPN target is:

$$\boxed{\alpha_1 = \alpha_2 = \alpha_3 = 0 \quad \textit{at the GR/local fixed point.}}$$

Large-scale medium corrections are allowed only through a reentry function $\mathcal{F}$ satisfying:

$$\boxed{\mathcal{F} \to 0 \quad \textit{in lab, Solar-System, binary-pulsar and local GW regimes.}}$$

This gate is open from $\mathcal{M}$; if it fails, flow/vorticity becomes an observable preferred frame.

Open Theorem 37 (MOND-safe reentry). *Derive a scale or acceleration gate:*

$$\boxed{\mathbf{a}_{\mathrm{med}} = \mathcal{F}\left(\frac{|\nabla\Phi|}{a_0}, \frac{L}{L_{\mathrm{bg}}}, \omega_{\mathrm{flow}}\right)\mathbf{a}_{\mathrm{flow}},}$$

with:

$$\boxed{\mathcal{F} \to 0 \quad \textit{in local tested regimes,} \qquad \mathcal{F} \neq 0 \quad \textit{in galactic low-acceleration regimes.}}$$

The exact function $\mathcal{F}$ is not fixed in RMG-1.

14 Strong-Field Compact-Object Contract

RMG-1 does not yet select the strong-field compact-object branch. The branch must be derived from the base-medium equilibrium law and the record/internal projection.

The currently allowed branches are:

$$\boxed{\text{GR/Kerr-like exterior,} \qquad \text{horizonless exponential branch,} \qquad \text{hybrid branch,} \qquad \text{alternative branch.}}$$

Candidate strong-field signatures — horizonless static exteriors, exponential freezing, curvature-soft deep boundaries, shifted ISCOs and breathing-mode tests — are listed as targets, not predictions. They become RMG-1 predictions only if they are derived from $\mathcal{M}$.

Gate 7 (Strong-field compact projection)**.** *For compact objects, the internal sector may dominate bound medium structure and contribute to total energy:*

$$\boxed{E_{\text{total}} = E_{\mathcal{R}} + E_{\mathcal{I}} + E_{\text{bind}}.}$$

However, the external observable cone must be defined by the record projection:

$$\boxed{\Pi_{\mathcal{R}} P_{\text{full}}(k) \Pi_{\mathcal{R}} = (G^{\mu\nu} k_\mu k_\nu)\, K_{\mathcal{R}},}$$

while internal compact modes may enter the gravitational source only through conserved total charges such as energy, angular momentum and allowed gauge charges. They must not define an independent external spacetime cone.

Open Theorem 38 (Hybrid compact-object program)**.** *Derive, or rule out, the hybrid strong-field branch:*

$$\boxed{\textit{external record geometry is GR/Kerr-like to current precision}}$$

while

$$\boxed{\textit{deep internal base-medium structure may freeze, soften or store energy.}}$$

The required proof is:

$$\boxed{\Pi_{\mathcal{I}} \textit{ contributes to } H_{\text{phys}} \quad \textit{but} \quad \Pi_{\mathcal{I}} \textit{ does not create a separate external cone.}}$$

If this fails, RMG must either reduce to a fully GR-like strong-field branch or derive and observationally confront an alternative strong-field branch.

15 Statement Status Map

This section is the integrated WP0 audit map. It is not a proof. It records which statements are postulates, which are conditional bridges, which are open theorems, and which gates block the first local-gravity MVP.

The sortable audit status vocabulary is restricted to

$$\boxed{\text{postulate, conditional result, derived result, target, open theorem, fail-risk.}}$$

Free-form context belongs in notes, not in the status label. Priority is used only for proof management:

$$\boxed{P0 = \text{MVP blocker,} \quad P1 = \text{first local package,} \quad P2 = \text{important later item,} \quad P3 = \text{extension window.}}$$

Foundation and local bridge.

- **RMG-001** one base-medium $\mathcal{M}$: postulate.
- **RMG-002** stable low-energy phase: P0 target. It is introduced as a primitive requirement, but the audit treats its physical realization as a condition to verify before Root A can be calculated.
- **RMG-004** Operational Relational Covariance: P0 target. It must be derived from microscopic record equivalence or replaced by a more explicit protection theorem.
- **RMG-005** positive physical energy: P0 target. It is introduced as a primitive requirement, but physical propagating modes still require a ghost/gradient-stability audit.
- **RMG-007–RMG-014** operational and weak-field bridge: conditional or derived local results. The weak bridge gives $u(r) = G_{\mathrm{RMG}}M/(rc^2)$ and $\gamma = \beta = 1$, but their final status remains conditional on Roots A–C.

Root blockers.

- **RMG-015** Root A: derive one external record carrier/cone from the stable phase. Blocks factorization, no-birefringence, the composite metric, and the one-cone part of Root B and Root C.
- **RMG-016** Root B: derive record-embedding ADM closure, including $\Gamma_{\mathcal{R}}$, $\Omega_{\mathcal{R}}$, embedding generators, ADM brackets, hyperbolicity and no ghost/extra-mode leakage. Blocks the ADM/EH fixed point and the two-TT-mode statement.
- **RMG-017** Root C: derive the total-energy source projection for all stable record matter. Blocks equivalence-principle source universality, compact-binary source tests and the no-fifth-force gate.
- **RMG-018** non-circular proof order: proof discipline target. Do not assume ADM to prove Root A, do not assume Einstein equations to prove Root B, and do not assume the equivalence principle to prove Root C.

Local-core gate clusters.

- **RMG-A-Factorization: RMG-019–RMG-023.** Prove $Q^{\mu\nu}_{AB} = G^{\mu\nu}K_{AB}$, eliminate the failure matrix $B^{\mu\nu}_{AB}$, forbid low-energy birefringence, remove extra observable rank-two tensors, and decouple local flow/vorticity from the tested PPN core.
- **RMG-B-ADM: RMG-024–RMG-027.** Prove $\Delta_{\mathcal{M}} = 0$ or suppression, close the hypersurface-deformation algebra, derive $\lambda = 1/2$, and obtain the ADM/EH fixed point.
- **RMG-GW-Tensor: RMG-028–RMG-030.** Prove exactly two positive, luminal transverse-traceless tensor modes and a detector-visible TT/shear response; forbid scalar/vector, massive spin-2, negative-energy or extra shear radiation in the tested local sector.
- **RMG-C-Source: RMG-031–RMG-037.** Prove $q_A \propto E_A/c^2$, the far-tail energy monopole, local $\sigma_{\mathrm{grav}} \propto \rho_E/c^2$, no species-private source, internal and binding-energy effacement, no species-private u-coupling, and matter-coupled ADM closure.

- **RMG-Micro: RMG-038–RMG-041.** Specify the microscopic input package, test the candidate order parameter $\sqrt{n}e^{i\theta}\chi$, derive $S^{\mathrm{rec}}_{\mathrm{eff}}[\mathcal{M}_0]$, and construct the composite record metric $g^{\mu\nu}_{\mathrm{rec}} = \mathcal{G}^{\mu\nu}[\Phi_{\mathcal{M}}]$.

Coverage rows added by the audit. The following rows are now tracked explicitly, even where the body of RMG-1 has not yet promoted them to standalone theorems:

- **RMG-045** GW emission and radiation reaction: derive tensor quadrupole energy loss, radiation reaction and compact-binary waveform phase from the same record metric and total-energy source. Propagation alone is not enough.
- **RMG-046** cosmological background reduction: specify the large-scale background state, expansion law and any dark-energy-like sector without using cosmology to cover a local-core failure.
- **RMG-047** finite strong-field tests: identify the ISCO, shadow, ringdown, strong-lensing and late-inspiral observables that must remain GR/Kerr-like or become explicit falsification targets.
- **RMG-048** quantum-extension boundary: state whether quantum claims are outside the MVP, imported as targets or derived later from the base-medium record action.
- **RMG-049** concrete Solar-System observation map: connect Newtonian limit, redshift, light bending, Shapiro delay, perihelion/ranging and frame-dragging tests to explicit RMG gates.

Observational sub-items.

- **OBS-001** Newtonian/orbital limit: checks RMG-012 and RMG-031–RMG-033.
- **OBS-002** gravitational redshift: checks RMG-009, RMG-010 and RMG-013.
- **OBS-003** light bending and Shapiro delay: checks RMG-008, RMG-013 and RMG-049.
- **OBS-004** perihelion, ranging and PPN β: checks RMG-013, RMG-027 and RMG-049.
- **OBS-005** frame dragging and preferred-frame null tests: check RMG-016, RMG-023, RMG-027 and RMG-049.
- **OBS-006** binary-pulsar orbital decay: checks RMG-017, RMG-031, RMG-035 and RMG-045; no leading scalar dipole loss is allowed.
- **OBS-007** GW propagation: checks RMG-028 and RMG-030; requires $c_{\mathrm{GW}} = c$ and exactly two tensor polarizations.
- **OBS-008** compact-binary waveform phase: checks RMG-028–RMG-030 and RMG-045; propagation and emission must be consistent.
- **OBS-009** strong-field finite tests: checks RMG-043, RMG-044 and RMG-047.
- **OBS-010** MOND transition: checks RMG-023 and RMG-042; any large-scale correction must not leak into the local core.
- **OBS-011** cosmological background: checks RMG-046.

For OBS-004 the chain must be explicit:

$$\boxed{\lambda = \frac{1}{2} \Rightarrow \text{first-class ADM closure} \Rightarrow \text{EH two-derivative local fixed point} \Rightarrow \text{nonlinear weak-field expansion} \Rightarrow \beta =}$$

If any arrow is imported rather than derived or audited, the corresponding observation row remains conditional.

Medium-to-target verification interface. The microscopic input package must be tested against the low-energy target by explicit checks: compute $\mathrm{Spec}(\mathcal{L}_{\mathcal{M}_0})$ and count the record cones; derive $\Pi_{\mathcal{R}}$ and $\Pi_{\mathcal{I}}$ and show that internal branches do not become private spacetimes; identify presymplectic null directions and first-class generators; compute the Schur-complement correction from internal integration and verify that it does not alter the local principal tensor or ADM bracket; compute the tensor kinetic rank, sign, mass and cone; and vary the lapse/metric to check that rest, kinetic, binding, internal/confined and field energy enter the same source. In particular, the chain

$$\boxed{\mathcal{D}_{\mathcal{M}} \to S_{\text{eff}}^{\text{rec}} \to S_{\mathcal{R}}[g_{\text{rec}}, \Psi_{\text{rec}}]}$$

must be explicit enough to run the Root B presymplectic, bracket and hyperbolicity audits.

MVP linearization scope. The first local-gravity MVP starts with a Minkowski local core for the principal-symbol, mode-count and hyperbolicity tests; a weak static local background is then used for the PPN/nonlinear-coefficient check; FLRW belongs to the cosmology window. The first pass requires the quadratic action in metric perturbations, the linearized constraints and their relevant quadratic-order closure, leading weak-field matter coupling, exactly two TT propagating modes, and a leading tensor-emission audit. Gauge fixing, such as harmonic gauge, may be chosen only after constraints are derived and may not hide kinetic lapse/shift or scalar/vector modes. Higher post-Newtonian terms and full nonlinear closure remain later targets.

RMG-distinctive observables. The protected local core is deliberately GR-like. RMG becomes observationally distinctive only if controlled windows produce falsifiable deviations after the local core is protected:

- high-precision delay tests may search for suppressed post-GR corrections only after Root A/B/C close;
- compact-binary waveform residuals, ISCO shifts, shadows and ringdown features belong to the compact-object branch;
- galaxy rotation belongs to MOND-like large-scale reentry and must not leak into Solar-System, binary-pulsar or local GW tests;
- cosmological acceleration belongs to the background-state closure, where Λ must be derived, regulated or replaced by a medium background law;
- preferred-frame null tests remain local-core failure tests, not deviation-window successes.

The immediate P0 blockers are therefore the stable phase, relational covariance/protection, positive-energy physical modes, the local effective hierarchy, Roots A–C, factorization, $\Delta_{\mathcal{M}} = 0$, ADM/EH closure, total-energy source universality, and the microscopic input package.

16 Open Items

RMG-1 is not yet a closed theory. The open items group into the following programs.

Microscopic foundation. Derive $S^{\rm rec}_{\rm eff}[\mathcal{M}_0]$ from the stable base-medium phase; specify the input package $\mathcal{D}_{\mathcal{M}} = (\Phi_{\mathcal{M}}, S_{\mathcal{M}}, \mathcal{M}_0, \ell_*, \Omega_{\mathcal{M}}, \mathcal{G}_{\rm int}, \mathcal{O}_{\rm rec})$; test the candidate order parameter $\Phi_{\mathcal{M}} = \sqrt{n}e^{i\theta}\chi$ against Roots A–C; compute the stable-phase spectrum; integrate out the internal sector and verify the absence of background leaks; derive ℓ_* from the base-medium record dynamics; verify that the stable phase contains no negative-energy physical branch; carry at least one working toy instantiation through its quadratic Hessian, spectrum, projectors and record/internal split.

Root A. Derive one common external record carrier from the stable phase; compute $\mathrm{Spec}(\mathcal{L}_{\mathcal{M}_0})$ and prove one gapless record branch; derive phase locking of stable tails to ν_0; prove the phase/acoustic branch gives only the common cone, not a scalar graviton; prove every long-range stable record family uses the coefficients $C_t : C_0 : C_x = 1 : \rho : \rho^2$.

Root B. Derive record-embedding first-class ADM closure; construct $\Gamma_{\mathcal{R}}$ and the presymplectic form $\Omega_{\mathcal{R}}$; derive the embedding vector fields δ_ξ and the local generators D, H; derive the Noether split $J_\xi = C_\xi + dQ_\xi$; verify B1–B5 and prove the anomaly gate $A[N, M] = 0$; derive the construction bridge $\mathcal{D}_{\mathcal{M}} \to S^{\rm rec}_{\rm eff} \to S_{\mathcal{R}}$; exclude unsuppressed lapse/shift scalar-vector kinetic terms as a derived Dirac result rather than as a gauge assumption; compute brackets from $\Omega_{\mathcal{R}}$ with Dirac brackets where needed; derive the hypersurface-deformation algebra; derive the DeWitt coefficient $\lambda = 1/2$; prove $\Delta_{\mathcal{M}} = 0$ or suppressed with explicit structural or scale tests; give a linearized hyperbolicity certificate, match the principal symbol to the record cone, and exclude ghost/gradient instabilities; verify constraint preservation under evolution and matter-coupled Dirac closure.

Root C. Verify C1–C5 in the base-medium matter sector; prove $S_u = \rho_E$ for every stable record matter family; prove metric-exhaustion of local u-dependence; prove the absence of explicit species-private u-coupling; prove $\sigma^{\rm long,local}_{\rm species} = 0$; derive the universal gravitational tail projection $q_A \propto E_A/c^2$ and the far-tail monopole $Q^{(0)}_A = \eta E_A/c^2$; audit compact-object sensitivities so that no leading scalar dipole loss appears in binary-pulsar systems; include self-gravitational binding energy in the strong-equivalence/Nordtvedt audit.

Factorization and protection. Derive $Q^{\mu\nu}_{AB} = G^{\mu\nu}K_{AB}$ from the record kinetic kernel; prove the stable record phase has only one observable principal tensor; derive non-common mode elimination; prove the observable tensor basis gate; eliminate unsuppressed $U^\mu U^\nu$ and flow/vorticity tensors; derive local flow decoupling and the preferred-frame PPN null target; prove no physical low-energy birefringence; derive one-cone clock universality; prove the failure matrix $B^{\mu\nu}_{AB}$ vanishes on physical principal modes.

Composite metric and tensor sector. Derive the composite map $g^{\mu\nu}_{\rm rec} = \mathcal{G}^{\mu\nu}[\Phi_{\mathcal{M}}]$; prove $\delta g_{\rm rec}$ contains a protected h^{TT}_{ij} sector; close the tensor kinetic-rank test (exactly two positive luminal TT modes and no extra tensor/scalar/vector radiation); prove amplitude and internal modes are gapped, constrained, gauge, confined or suppressed.

Radiation and detection. Separate propagation from emission. For propagation, derive scalar/vector constraints and tensor/shear radiation; derive lapse/shift no-kinetic structure from $\mathcal{M}$; prove first-class constraint closure removes scalar/vector radiation; prove pure common-mode breathing is locally self-calibrated and detector-null; derive the detector-visible TT/shear or defect projection for GW signals. For emission, derive tensor quadrupole energy loss, radiation reaction and compact-binary waveform phase from the same record metric and total-energy source. Exclude massive vector modes, massive spin-2 leakage, a second metric tensor mode and any hidden scalar/vector energy-loss channel in the tested local sector.

Observational audit package. Map each local observable to a gate: Newtonian/orbital limit; gravitational redshift; light bending and Shapiro delay; perihelion, ranging and PPN β; frame dragging and preferred-frame null tests; binary-pulsar orbital decay; GW propagation; compact-binary waveform phase; and finite strong-field tests such as ISCO, shadow, ringdown, strong lensing and late-inspiral behavior. Separate GR-like local-core recovery from RMG-distinctive deviation-window observables.

Foundational and extension programs. Derive Operational Relational Covariance from microscopic $\mathcal{M}$; derive the record/internal projection $\Pi_{\mathcal{R}}$ from $\mathcal{M}$; derive the flow/vorticity gate and the MOND reentry function $\mathcal{F}$; derive the compact-object branch and the strong-field projection gate; derive the cosmological background state and the matter sector from $\mathcal{M}$; state the quantum-extension boundary, namely which quantum claims are outside the local-gravity MVP and which must be derived later from the base-medium record action.

17 Pass/Fail Gates

Gate 8 (Pass condition). *The framework passes if $\mathcal{M}$ derives $S_{\rm EH} + S_{\rm matter}$ as the low-energy fixed point, with positive physical Hamiltonian, universal metric coupling, and exactly two vacuum tensor gravitational degrees of freedom.*

Gate 9 (Fail conditions). *The framework fails in the tested local sector if any of the following occurs:*

- *observable preferred frame, species-dependent metric coupling, extra scalar/vector radiation, wrong gravitational-wave speed, negative-energy physical modes, or non-closing ADM constraints;*
- *factorization failure $Q^{\mu\nu}_{AB} \neq G^{\mu\nu} K_{AB}$, with the failure matrix $B^{\mu\nu}_{AB}$ producing species-dependent cones, a preferred-frame tensor, or a geometry cone distinct from the matter cone;*
- *composition-dependent gravitational projection $q_A/(E_A/c^2)$, or long-range tails requiring distinct spacetime carriers;*
- *independent $u^\mu_{\mathcal{M}}$ in local operators, or low-energy kinetic terms for lapse or shift;*
- *residue $\Delta_{\mathcal{M}}$ leaving an unsuppressed preferred-frame term, extra operational tensor, non-ADM constraint bracket, or scalar/vector propagating mode;*
- *observed interferometer signal explained only by pure common-mode breathing that should be self-calibrated, or unsuppressed breathing/scalar polarization in local polarization tests;*
- *record dynamics that do not yield exactly two positive, luminal, common-cone TT modes, or that leave an unsuppressed scalar/vector, massive spin-2, negative-energy or extra shear channel;*
- *binary-pulsar timing or compact-binary waveforms requiring an energy-loss law different from the protected tensor channel, or requiring an unsuppressed scalar dipole/monopole radiation channel;*

- *unsuppressed MOND-type correction in solar-system, binary-pulsar or local GW tests, or modification of $\gamma = 1$, $\beta = 1$, $c_{\rm GW} = c$, universal free fall, or the two-tensor-mode vacuum sector;*

- *compact-object sector hiding energy from gravity, creating an unsuppressed breathing/scalar radiation channel, defining a species-private external cone, altering $c_{\rm GW} = c$, or forcing shadow, ringdown, binary-pulsar or strong-field lensing deviations already excluded by observation.*

Current Status

$$\boxed{\text{RMG-1 has a conditional leading-order local GR core, not yet a closed theory.}}$$

Principal theorem to be derived:

$$\boxed{\mathcal{M} \Rightarrow \text{exact ORC/no-extra-tensor} \Rightarrow Q^{\mu\nu}_{AB} = G^{\mu\nu}K_{AB} \Rightarrow \Delta_{\mathcal{M}} = 0 \Rightarrow \text{ADM/Einstein-Hilbert low-energy fixed poi}}$$